Department of
CIVIL AND ENVIRONMENTAL ENGINEERING
AT NORTHEASTERN UNIVERSITY

# CIVE 2261

# Materials for the Built Environment

## *Laboratory Manual*

*Revised 2023*

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*Companion Labs to* **CIVE 2261: Materials for the Built Environment**

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# Preface

*"In the spirit of science, there really is no such thing as a 'failed experiment.' Any test that yields valid data is a valid test."*

Adam Savage, *Mythbuster*

## Intended Laboratory Manual Use

This laboratory manual is intended to be used across one semester in the course **CIVE 2261: Lab for CIVE 2260** offered by the Department of Civil and Environmental Engineering at Northeastern University in Boston, Massachusetts. This course is intended to be taken concurrently with **CIVE 2260: Materials for the Built Environment**. The laboratory exercises provide hands-on, experiential understanding to complement the theory and concepts discussed in **CIVE 2260**.

Details neglected herein are assumed to be presented either in class or in lab via lecture.

Note also that the labs listed herein are subject to revision at the behest of the instructor/professor.

## Laboratory Overview

In this class, you will be introduced to standard laboratory procedures for materials testing, data reduction and data analysis methods.

This manual is comprised of five laboratories; however, the instructor/professor may adjust this number. Laboratory methods and determinations covered in this manual include: uniaxial tension, concrete mixing and testing, fracture analysis, and flaw tolerance. The operation and use of computer-based test control and data acquisition systems are also covered.

Each lab will consist of a short lecture/introduction, conducting a lab test (or tests), and clean-up. The labs should be readily completed within the allotted time period.

Currently, the labs are designed for group work and group lab reports. Confirm with your instructor/professor.

By participating in this laboratory and completing the assignments, you will gain experience preparing laboratory reports using effective technical writing skills, including graphic presentation of data.

## Laboratory Objectives

The primary objectives of the Materials for the Built Environment Laboratory are as follows:

1. Apply accepted standards (i.e., ASTM Standards) related to laboratory testing of materials to determine mechanical properties.
2. Demonstrate knowledge of the basic principles of stress and strain and how they relate to test results and overall material performance.
3. Explain different failure modes (e.g., brittle vs. ductile, fracture), flaw tolerance and material toughness.

4. Explain how to design concrete mixes for particular applications, including the effect of water content.
5. Explain how to mix concrete and prepare specimens for strength testing.
6. Write effective technical reports to communicate the results of their laboratory testing, including preparing effective graphics.

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# Laboratory Safety and Safety Training Assignment

# Laboratory Safety and Safety Training Assignment

All Safety Information Must be Reviewed and Trainings Completed Before Participating in Lab Activities

## Introduction

Safety is a collective responsibility that requires the full cooperation of everyone in the laboratory. However, the ultimate responsibility for safety rests with the person carrying out a given procedure. Accidents often result from an indifferent attitude, failure to use common sense, or failure to follow instructions. Each student should be aware of what the other students are doing because all can be victims of one individual's mistake. Do not hesitate to point out to other students that they are engaging in unsafe practices or operations. If necessary, report it to the instructor.

The following are general guidelines for all laboratory workers:

1. Follow all safety instructions carefully.
2. Become thoroughly acquainted with the location and use of safety facilities such as safety showers, exits, extinguishers, and eyewash stations.
3. Become familiar with the hazards of the chemicals or equipment being used and know the safety precautions and emergency procedures before undertaking any work.
4. Become familiar with the procedures and hazards involved before beginning an experiment.
5. The laboratory is a place for serious learning and working. Horseplay cannot be tolerated.
6. Avoid working alone in a building or in a laboratory.

## Laboratory Safety Requirements

Prior to working in the lab (attending lab and conducting any of the lab tests/experiments), all students must complete safety training. Students can demonstrate they have completed the training by showing their certificate of completion for ALL required trainings. Any students who have not completed ALL of the training before coming to work in lab will not be permitted to attend the lab. The mandatory training courses required by university policy can be accessed at https://northeastern.scishield.com/ (then click on 'Log in with you Northeastern UserID and Password'). The mandatory training courses include the following:

- Orientation for Labs, Makerspaces and Studios is **MANDATORY** for all lab users. This is a one-time online training
- Fundamentals of Laboratory Safety is **MANDATORY** for all lab users. This is a one-time online training with a required refresher every year

Keep in mind that some of the trainings require periodic refreshers and you should keep them up to date as long as you are an active laboratory user.

## Personal Protection

Everyone working in the lab must wear the proper protective equipment. In this case, the following items are required of all persons in the laboratory:

- Protective eyewear (safety glasses)
- Proper clothing (long pants, closed toe shoes)
- Lab coats
- Gloves (when using chemicals or mixing concrete)

Any student who does not arrive at the lab with the proper attire or refuses to wear the proper safety attire must be asked to leave. THIS POLICY WILL BE ENFORCED. In addition, per policies, food and beverages are NOT PERMITTED in the laboratory.

### *Protective eyewear*

All people in the laboratory, including visitors, must always wear eye protection, even when not performing an experiment or chemical operation. Wearing of contact lenses in the laboratory is normally forbidden because contact lenses can hold foreign materials against the cornea. Furthermore, they may be difficult to remove in the case of a splash. Soft contact lenses present a hazard because they can absorb and retain chemical vapors. If the use of contact lenses is required for therapeutic reasons fitted goggles must also be worn. Normal prescription eyeglasses, though meeting the Food and Drug Administration's standards for shatter resistance, do not provide appropriate laboratory eye protection.

### *Clothing*

Clothing worn in the laboratory should offer protection from splashes and spills, should be easily removable in case of accident, and should be at least fire resistant. Lab coats provided by the NU lab coat program meet these requirements and must be worn when working in the laboratory.

High-heeled or open-toed shoes, sandals, or shoes made of woven material should not be worn in the laboratory. Shorts, cutoffs and miniskirts are also inappropriate. Long hair and loose clothing should be constrained. Jewelry such as rings, bracelets, and watches should not be worn to prevent chemical/water seepage under the jewelry, contact with electrical sources, catching on equipment, and damage to the jewelry.

### *Gloves*

Gloves can serve as an important part of personal protection when they are used correctly. Check to ensure the absence of cracks or small holes in the gloves before each use. Gloves should be removed before leaving the work area and before handling such things as telephones, doorknobs, writing instruments, computers, and laboratory notebooks.

### *Personal Hygiene*

Everyone working in an academic laboratory should be aware of the dangers of ingesting/ inhaling chemicals. These common-sense precautions will minimize the possibility of such exposure:

1. Do not prepare, store (even temporarily), or consume food or beverages in any laboratory.

2. Smoking is strictly prohibited in any of the university premises. Additionally, be aware that tobacco products in opened packages can absorb chemical vapors.
3. Do not apply cosmetics in a laboratory.
4. Wash hands and arms thoroughly before leaving the laboratory, even if gloves have been worn.
5. Wash any piece of clothing on which chemicals have been spilled separately from personal laundry.
6. Never wear or bring lab coats into areas where food is consumed.

## Laboratory Protocol

### *Housekeeping*

In the laboratory and elsewhere, keeping things clean and neat generally leads to a safer environment. Avoid unnecessary hazards by keeping drawers and cabinets closed while working. Never store materials, especially chemicals, on the floor, even temporarily. Work spaces and storage areas should be kept clear of broken glassware, leftover chemicals/materials and scraps of paper. Keep aisles free of obstructions such as chairs, boxes and waste receptacles. Avoid slipping hazards by keeping the floor clear of small items and spilled liquids. Contact the instructor to follow the required procedure for the proper cleaning of spills and or disposal of wastes that require a special treatment.

### *Cleaning Glassware*

Clean glassware at the laboratory sink or in laboratory dishwashers. Use hot water, if available, and soap or other detergent. If necessary, use a mild scouring powder. Wear appropriate gloves that have been checked to ensure that no holes are present. Use brushes of suitable stiffness and size. Avoid accumulating too many articles in the cleanup area. Usually work space around a sink is limited and piling up dirty or cleaned glassware leads to breakage. Remember that the turbid water in a sink may hide a jagged edge on a piece of broken glassware that was intact when put into the water. A pair of heavy gloves may be useful for removing broken glass, but care must be exercised to prevent glove contamination. To minimize breakage of glassware, sink bottoms should have rubber or plastic mats that do not block the drains.

### *Unattended Operation of Equipment*

Experiments that are left to run unattended are prime sources for accidents. Do not let equipment run overnight without fail-safe provisions and the instructor's consent. Always leave a note plainly posted with a phone number where you and the instructor can be reached in case of emergency. Remember that in the middle of the night, emergency personnel are entirely dependent on accurate instructions and information.

## Use of Chemicals

Before using any chemical, you need to know how to safely handle it. The safety precautions taken are dependent on the exposure routes and the potential harmful effects. There are four routes of exposure:

ingestion, inhalation, absorbed through skin, and through eye contact. Each potential exposure route requires different precautions.

Chemical exposure may have acute (immediate, short term) or chronic (long term potentially cumulative) affects. Information on health hazards can be found on chemical labels and in Safety Data Sheets (SDS).

SDS for most chemicals used in the laboratory are located on the bookshelf at the entrance of the laboratory. Electronic versions (potentially more current) can be found on the Internet. SDS provide extensive information on safe handling, first aid, toxicity, etc.

Chemicals or materials that can potentially cause a hazard are rated by the National Fire Protection Association (NFPA). Ratings are included to indicate the types and severity of the hazards. The NFPA ratings are on a scale of 0-4 with 0 being nonhazardous and 4 being most hazardous. The ratings are described in **Table 0.1**.

**Table 0.1: NFPA hazard code ratings.**

| Code | Health | Fire | Reactivity |
|---|---|---|---|
| **4** | Very short exposure can cause death or major residual injury | Will rapidly or completely vaporize at normal pressure and temperature | Capable of detonation or explosive reaction at normal temperatures and pressures |
| **3** | Short exposure can cause serious temporary or residual injury | Can be ignited under almost all ambient temperatures | Capable of detonation or explosive reaction buy requires a strong initiating source or must be heated under confinement before initiation |
| **2** | Intense or continued exposure can cause temporary incapacitation or possible residual injury | Must be moderately heated or exposed to high temperature before ignition | Undergoes violent chemical change at elevated temperatures and pressures or reacts violently with water. |
| **1** | Can cause irritation but only minor residual injury | Must be preheated before ignition | Normally stable but can become unstable at elevated temperatures and pressures. |
| **0** | During a fire offers no hazard beyond combustion | Will not burn | Stable even under fire conditions. |

### *Chemical Storage*

There has been much concern, and some confusion, about the proper storage of laboratory chemicals. Here “proper” means the storage of chemicals in such a manner as to prevent incompatible materials from being accidentally mixed together in the event of the breakage of one or more containers in the storage area or to prevent the formation of reactive vapors that may require vented chemical storage areas. Even if you are not working with any chemicals or materials that can potentially cause a hazard, check the following guide to the storage of common laboratory chemicals that may be present in the lab.

1. Perchloric acid is separated from all other materials.
2. Hydrofluoric acid is separated from all other materials.
3. Concentrated nitric acid is separated from all other materials.
4. Highly toxic materials ($LD_{50}$ of 50 mg/kg or less) are stored separately.
5. Carcinogenic chemicals are stored separately.
6. Inorganic acids (except for 1, 2, 3 above) are stored separately.
7. Bases are stored separately.
8. Strong oxidizing agents are stored separately.
9. Strong reducing agents are stored separately.
10. Water reactive, pyrophoric and explosive materials are stored separately.
11. Flammable organic materials (solvents, organic acids, organic reagents) are stored separately.

In order to minimize the generation of waste materials, reaction byproducts and surplus chemicals will be neutralized or deactivated as part of the experimental procedure. Waste materials must be handled in specific ways as designated by federal and local regulations. Some general guidelines are:

1. Dispose of waste materials promptly. When disposing of chemicals, one basic principle applies: Keep each different class of chemical in a separate clearly labeled disposal container.
2. Never put chemicals into a sink or down the drain unless they are deactivated or neutralized and unless they are allowed by local regulation in the sanitary sewer system.
3. Put ordinary waste paper in a wastepaper basket separate from the chemical wastes. If a piece of paper is contaminated, such as paper toweling used to clean up a spill, put the contaminated paper in the special container that is marked for this use. It must be treated as a chemical waste.
4. Broken glass belongs in its own marked waste container Broken thermometers may contain mercury in the fragments and these belong in their own special sealed "broken thermometer" container.

For any additional information please refer to the CEE laboratory safety plan available at: https://cee.northeastern.edu/wp-content/uploads/sites/3/pdfs/ceelsp.pdf

## References

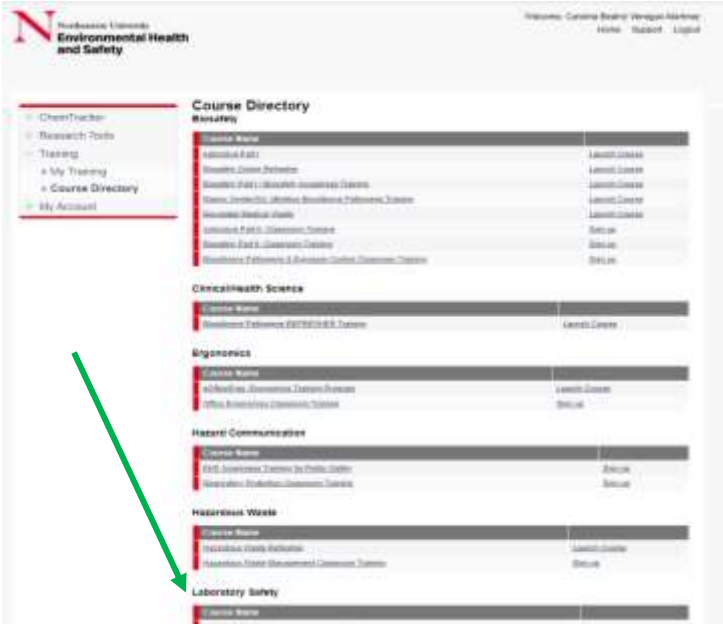
1. Safety in Academic Chemistry Laboratories. A publication of the American Chemical Society Committee on Chemical Safety. Fifth edition. 1990
2. OSHA Laboratory Standard
3. 2019 CEE Laboratory Safety Plan. Northeastern University.

# Laboratory Safety Training Assignment

For using the Snell 068 lab, each individual needs to pass the two training courses that are listed below and fill out the "CEE Laboratory User Form." These training courses are mandatory for all lab users due to university safety policies. If you have taken these trainings less than 1 year ago, you are exempted from taking them again, but you do need to submit your completion certificates.

The trainings can be accessed here: https://northeastern.scishield.com/, by clicking on 'Log in with your Northeastern UserID and Password'), logging in with your NU credentials, and clicking on the "Training" link on the menu at the left hand side of the webpage. To locate the trainings, click on "Course Directory" and then scroll down to the "Laboratory Safety" section.

- Orientation:
  - "Orientation for Labs, Makerspaces and Studios" is **MANDATORY** for all lab users. **It is a one-time online training**
- Laboratory Safety/Chemical Hygiene:
  - "Fundamentals of Laboratory Safety" is **MANDATORY** for all lab users. **It is a one-time online training** + refresher every year (online)




To obtain your certificate for completed training, navigate to the "Training" link in the menu at the left side of the SciShield webpage, then click "My Training." You will see a list of all completed trainings, with a link to access the certificate. After taking the training successfully, download and save your certificate of completion for each training course. Combine both certificates, along with the completed CEE lab user form (see next page) into one pdf file (e.g., a 3-page pdf, 1 page for each certificate and 1 page for the lab user form) and submit the combined file by uploading it on the assignments section of Blackboard, as indicated by the instructor.

Students who do not submit the assignment are not allowed to attend the first lab and will be considered absent.

# CEE Laboratory User Form

As a laboratory user in the Department of Civil and Environmental Engineering at Northeastern University, I confirm that I have read – and will comply with – the Department's Laboratory Safety Plan, which is available at https://cee.northeastern.edu/wp-content/uploads/sites/3/pdfs/ceelsp.pdf

I understand that if I am found in noncompliance with any provisions of the Laboratory Safety Plan that my laboratory privileges may be suspended or revoked.

This form must be signed, dated, and returned to the instructor before any work is conducted in any department laboratory.

| | |
|---|---|
| Print Name | Signature |
| NU ID# | Date |

# Laboratory #1: Uniaxial Tension

# Laboratory #1

Uniaxial Tension

## Objectives and Overview

The uniaxial tension test is one of the most common tests in materials science. Simply put, a material specimen is placed under a state of increasing stress until it fails. Tension tests provide information on the strength and ductility of materials under uniaxial tensile stresses. Properties that are directly measured include ultimate tensile strength, and maximum elongation (ultimate strain). Analysis of the material behavior can yield other properties including Young's modulus (E) and Poisson's ratio ($\nu$). A uniaxial test is most applicable for isotropic materials.

During this lab, you will perform tension tests on *both* metallic and plastic specimens. Testing will be conducted on specimens of varying strengths and behaviors (steel, aluminum, PVC) to illustrate material differences. For the experiment, a series of specimens will be loaded in tension up to failure, and thus the test is known as a destructive test (you cannot use the specimens more than once!).

The instrument used for the test records force-displacement (F-δ) data that will allow us to plot stress-strain (σ-ε) curves for each specimen, giving us insight into the mechanical behavior of the material. From the stress-strain curve, we will be able to determine properties such as yield strength ($\sigma_y$) and ultimate strength ($\sigma_{max}$).

## Definitions

- **Brittle Materials:** Brittle materials have a small percent reduction in area, and the fracture stress is the same as the ultimate stress. They also have small deformations.
- **Ductile Materials:** Ductile materials have the ability to permanently deform without fracture (plastic deformation). The ductility of a material is described by percent elongation and percent area reduction.
- **Elastic Deformation:** Recoverable deformation; limit at which release of load results in zero permanent strain.
- **Modulus of Elasticity (Young's Modulus; *E*):** Slope of the σ-ε diagram at any point (see **Figure 1.1**). For linear elastic behavior, Young's modulus is constant. For nonlinear behavior, the value of Young's modulus changes depending on the amount of stress or strain.
- **Modulus of Resilience:** Capacity to absorb energy *without* permanent deformation. Equal to the area under the elastic portion of the σ-ε curve.
- **Modulus of Toughness:** The amount of energy required to fracture a specimen. Equal to the total area under the σ-ε curve.
- **Offset Yield Strength:** Yield strength defined by a specified offset in strain. When the yield strength is not well defined, the offset yield strength is calculated. Offsets vary by type of material (i.e., steel – 0.2%, gray cast iron – 0.05%, aluminum alloys – 0.2%, brass – 0.35%).

For example: 0.2% offset – locate on the strain ordinate the value 0.002 in/in. Draw from this point a line parallel to the linear portion of the σ-ε diagram. The intersection of this line with the σ-ε curve is the yield strength at 0.2% offset strain.

- **Percent Elongation:** Total percent increase in length of specimen during tensile test.
- **Plastic Deformation:** Unrecoverable deformation; limit at which release of load results in some permanent strain.
- **Proportional Limit ($\sigma_p$):** Point at which the σ-ε curve becomes non-linear (see **Figure 1.1**).
- **Rupture or Fracture Strength:** Applied stress at failure; e.g., the stress when maximum strain is attained; not necessarily peak stress (see **Figure 1.1**).
- **Strain (ε):** Relative axial deformation; the change in length divided by the original length of the specimen.
- **Strain Hardening:** After the yield point, the specimen requires an increase in stress in order to strain further. On a Stress-Strain curve, strain hardening is when there is a positive slope between the yield point and ultimate strength.
- **Ultimate Tensile Strength ($\sigma_{max}$):** The stress on a material under the maximum tensile force/load (see **Figure 1.1**).
- **Yield Point:** Most commonly, the stress at which an increase in strain occurs without an increase in stress (e.g., a plateau in the σ-ε diagram). At this point, permanent plastic deformation is commonly initiated (see **Figure 1.1**).

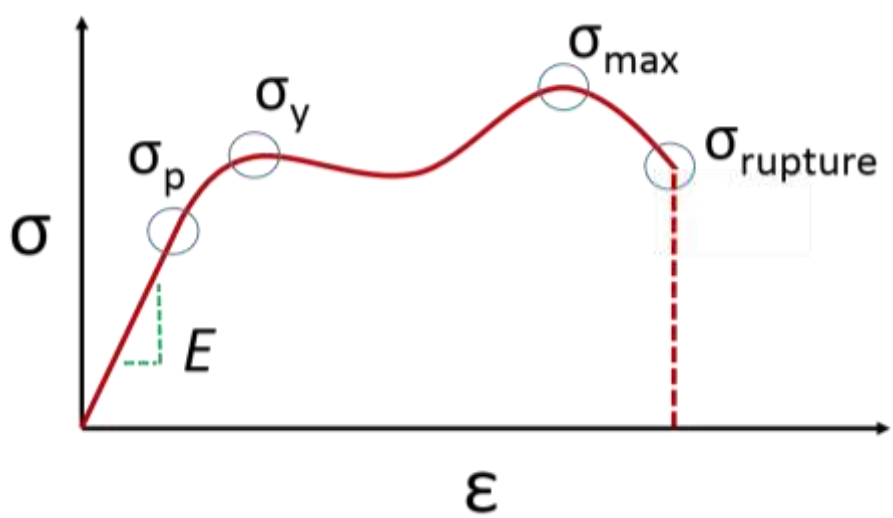



**Figure 1.1: Example stress-strain (σ-ε) plot indicating key stress points and features, including Young's modulus, proportional limit ($\sigma_p$), yield strength ($\sigma_y$), ultimate tensile strength ($\sigma_{max}$), and rupture strength ($\sigma_{rupture}$). Note that the features in the above plot are exaggerated for illustrative purposes. Lab recorded results may not have clearly defined points.**

## Theory

**The required theory/data analysis for this lab includes:**

**(1) The conversion from force-displacement data to stress-strain data**

Stress is calculated by dividing the tensile force applied to the specimen by the original area of the specimen, or:

$$\sigma = \frac{F}{A_0}$$

where $\sigma$ is the stress (psi or MPa), $F$ is the load carried by the specimen during tension test (lb or N), and $A_0$ is the initial or original cross-sectional area of the specimen (in$^2$ or m$^2$).

Strain is calculated by dividing the change specimen length ($\Delta L = L - L_0$) by the original specimen length ($L_0$). However, the test equipment records the displacement (δ). The new specimen length is thus:

$$L = L_0 + \delta$$

Such that the strain is:

$$\varepsilon = \frac{\Delta L}{L_0} = \frac{L - L_0}{L_0} = \frac{(L_0 + \delta) - L_0}{L_0} = \frac{\delta}{L_0}$$

where $\varepsilon$ is the strain (in/in or m/m; e.g., unitless), $\delta$ is the recorded machine displacement (in or m), and $L_0$ the initial or original gage length of the specimen (in or m)

Thus, the output force-displacement data is easily converted to stress-strain data by recording the initial dimensions of the specimen.

**(2) The determination of Young's modulus via linear regression**

For linear elastic materials, stress is related to strain via Hooke's law, where:

$$\sigma = E\varepsilon$$

Note that this is just a linear equation, $y = Mx + b$, where $y$ is stress, $x$ is strain, the intercept is zero and the slope is Youngs' modulus, $E$. Note that this is equivalent to defining the modulus as:

$$E = \frac{d\sigma}{d\varepsilon}$$

Thus, to determine the modulus, one must simply fit a linear trendline to the stress-strain data.

Choose a section of the data that looks linear (typically for strains less than 2% or 3%, but is dependent on the material; use your own judgement!), and fit a trendline to it using Excel or Matlab.

Moreover, it is good practice to calculate the $R^2$ value, which will give an indication of the goodness of fit.

**(3) The determination of yield stress via the offset method**

The yield stress defines the onset of plastic strain. At a given stress, the material *yields* to the stress, and loses any additional capacity. For many materials, this is apparent due to a point of zero stiffness – e.g., it takes no increase in stress to deform a specimen further. Mathematically,

this occurs when the tangent modulus approaches zero (e.g., a horizontal slope). However, this is not always the case.

It is clear that yield involves a change in stiffness. For many materials without a clear yield point, we can simply offset the initial modulus. If no yield ever occurs, this offset would never intersect with the stress-stain curve.

However, when the stiffness decreases (due to yielding), the offset will result in the offset line intersecting the stress-strain curve. The stress at which this occurs is then approximated as the yield stress. Since the amount of offset is somewhat arbitrary, it must be reported with the yield stress (e.g., "*yield stress of 100 MPa at 0.1% offset*").

Since the Young's modulus has already been determined, the offset method simply shifts the tangent line horizontally to the necessary offset (say, 0.2%) and extends the fitted line until it intersects the stress strain plot.

To accomplish this, the equation of the offset must be determined. First, we know that, *via* Hooke's law: $\sigma = E\varepsilon$. To shift this by an offset, *s*, we need to introduce an intercept, such that:

$$\sigma_{offset} = E\varepsilon + b$$

The offset is on the horizontal (x-axis), while the intercept is on the vertical (y-axis), but we know the slope (modulus, *E*), such that:

$$E = \frac{dy}{dx} = \frac{b}{s} \rightarrow b = Es$$

Thus,

$$\sigma_{offset} = E\varepsilon + Es = E(\varepsilon + s)$$

One can easily plot this new equation by adding the offset to the strain values, and determining approximately where the intersection occurs.

**(4) The calculation of resilience/toughness moduli via numerical integration**

Both resilience and toughness are defined by areas under the stress-strain curves.

There are many methods of numerical integration that you may have learned in Calculus classes (e.g., Trapezoidal method, Simpson's Rule, etc.). Here, we apply one of the simplest methods – the Rectangle method.

In effect, we assume each of our data points represents a "rectangular area" of the stress-strain curve. We know the height of each rectangle (the stress value) and we know the width of each rectangle (the difference in adjacent strain values). Thus, the area up to any strain is just the sum of the rectangles up to that strain, where:

$$A = \sum_i \sigma_i \Delta\varepsilon_i = \sum_i \sigma_i(\varepsilon_{i+1} - \varepsilon_i)$$

Note that the rectangular method has significant error if large strain steps are used. However, the smaller the step size ($\Delta\varepsilon_i$), the smaller the error of the method. *E.g.,* as $\Delta\varepsilon_i \rightarrow 0$, the approximation approaches the exact value of the integral.

## Apparatus

This lab test consists of the following materials:

1. **Tension machine (ADMET UTS):** Your lab instructor will explain how the compression/tension machine works. In general, a specimen is stretched at a prescribed displacement rate (e.g., displacement controlled) and the load is recorded until the specimen fails.
2. **Digital calipers:** For measuring the length and diameter of the specimens.
3. **Digital scales:** For measuring the weight of the specimens.
4. **Three (3) material specimens:** (1) steel, (1) aluminum, and (1) PVC. The specimens are cylindrical rods. Depending on the capacity of the ADMET machine you use, the rods may or may not be notched at the center to reduce the cross-sectional area (and help initiate failure).
5. **Camera (*not provided*):** To take pictures of the specimens during testing and post-failure.

## Procedure

You will use the following procedure to conduct the tension testing.

1. Put the specimen into the grips on the ADMET machine. The grips should have V-notches to accommodate the cylindrical rods.
2. Measure the gage length of the specimen (the length over which the specimen can fail); e.g., the free length between grips. Measure the diameter of the specimen within the gage length. Record your data.
3. Verify the test settings, including displacement rate and failure conditions. Record these conditions. The displacement rate should be set to 1 inch/minute for the test.
4. Start the test.
5. Readings of applied force and deformation (elongation) are observed on the computer monitor, as well as the force-deformation curve.
6. At least one group member is responsible for taking pictures prior to the test, during the test, and after failure. These pictures should be included in the lab report as support for your analysis.
7. After failure, remove the specimen from the grips and inspect it.
8. The data acquisition system attached to the testing machine writes records of applied force and deformation at regular intervals to a file. Students will be given that file of force-deformation records (the format is delimited text, which can be opened in Excel or Matlab).
9. Repeat steps 1-8 for each material specimen.
10. Carry out the analysis described below.
11. Write your lab report, addressing all items in the Analysis section.

## Analysis

Your lab reports must contain the following:

1. Convert the computer readings (force-deformation) of the metallic and plastic specimens to stress-strain, and develop stress-strain plots for each of the three specimens (total of 3 plots).
2. On your plots, identify all of the important values (i.e., the values that you are reporting in the next analysis item #3).
3. For all metallic and plastic specimens, report the following in a table:
    a. Proportional Limit
    b. Yield Strength (use the offset method if yield point is not well defined)
    c. Ultimate Strength
    d. Specific Strength
    e. Rupture Strength
    f. Modulus of Elasticity
    g. Modulus of Resilience
    h. Modulus of Toughness
    i. Percent Elongation
    *j.* Type of Failure (brittle vs. ductile).

    *Note: material behavior can vary widely and the judgement between "ductile" and "brittle" failure behavior is not always obvious like it is when comparing the failure A36 steel and concrete, for example. Consider all aspects of the failure, such as strain hardening/softening, elongation and area reduction when making a judgement about the type of failure observed.*

## To Include in Results/Analysis/Discussion

1. Discuss and compare the failure of the tensile specimens.
2. Which material has the ability to absorb the most energy without permanent deformation? Explain why.
3. Calculate the percent reduction in area for each of the specimens, and use these findings to make a comparison between the different specimens.
4. What is strain hardening? Do any of the plots exhibit strain hardening?
5. What were the expected results/ranges (look up common values) for these materials? How do your results compare to these expectations?

## Reflection/Questions

1. The stress here is defined as the applied force over the original specimen area. This is commonly known as the “engineering stress”. How does the engineering stress differ from the “true stress” experienced by the specimen? How would it change your results, if at all?
2. The ADMET machines we use in the lab are displacement-controlled; that is, they stretch the material with a known displacement rate (such as 0.1 inches/minute) and the load is measured. Uniaxial tension tests can also be force-controlled; that is, stretching a material with a known loading rate (such as 100 lbs/minute). If the tests were force-controlled, would the results be the same? Why or why not?
3. Uniaxial tensile tests are most commonly performed for isotropic materials. Why?

# Laboratory #2: Concrete Mix

# Laboratory #2

Concrete Mix

## Objectives and Overview

*Note: It is likely that you are performing this lab without having covered concrete in class. This is due to the fact that concrete takes one month (28 days) to sufficiently cure. As such, specimens must be prepared in order to test them later in the semester.*

Concrete is a custom-designed product whose properties depend on the properties and relative proportions of its ingredients. Users of concrete care mostly about two properties: strength (after it is cured) and workability (before it is cured). The primary measure of strength is called compressive strength, for which we use the symbol $f_c'$; the primary indicator of workability is called *slump*.

In this lab, you will learn:

1. How to design a concrete mix to meet strength and workability specifications;
2. How to mix concrete;
3. How to test a batch of freshly mixed concrete to see if it meets its workability specification;
4. How to prepare concrete specimens (cylinders and beams) for testing to see whether the concrete meets its strength specification after it has cured.

Extensive experimentation by the American Concrete Institute (ACI) has led to design procedures that indicate the proportions of ingredients needed to achieve various specifications. The mix design used in this lab is based on the Absolute Volume Method, where the sum of the absolute volumes of the water, cement, aggregate, air and admixtures (used in some cases) is equal to the volume of the freshly mixed concrete. After you measure out the ingredients of the prescribed mix design, you will mix a small batch of concrete. You will learn about manually mixing concrete, including how to clean up afterwards.

You will test your concrete for workability using a slump test (ASTM procedure C143). The slump test is the most common test performed on wet concrete, and is simple enough to be carried out on site during construction projects.

Finally, you will form specimens of your concrete in cylindrical molds (ASTM procedure C31) to be used for strength testing after the concrete has cured. Concrete cylinders are typically formed for every batch of concrete; testing usually occurs after curing for 28 days.

Typically, if the concrete specimens fail to meet the desired test strength, the project will have to come to an abrupt halt while engineers determine whether the deficient concrete has to be replaced or otherwise compensated for – a very expensive proposition. This kind of testing keeps concrete mixers honest. If their batch was not being tested, they could save a lot of money by adding more water, more sand, and less cement.

## Definitions

- **Advantages of Using Concrete:** High compressive strength, non-combustible, durable, abrasion resistant, practically impermeable, requires little maintenance, and can easily be formed into a variety of shapes.
- **Disadvantages of Using Concrete:** Concrete is very weak in tension (has approximately 10% of compressive strength in tension).
- **Bleeding:** Migration of water to the surface of the concrete caused by settlement of solid materials. Excessive bleeding causes the w/c ratio to increase greatly at the surface, creating a weak layer that can result in poor durability of the concrete.
- **Compressive Strength ($f_c'$):** The common measure of a concrete specimen's strength. Most "general use" concrete has a compressive strength in the range of 3,000 - 5,000 psi; however, strengths as low as 2,000 psi and as great as 20,000 psi (high strength concrete) are sometimes used, depending on the application.
- **Concrete Components:** Concrete mixtures consist of two main components: paste and aggregate.
    - ***Paste*:** is made up of Portland cement, water, and air, and accounts for approximately 25-40 percent of the total volume. The paste binds the aggregate into a rock-like mass when it hardens due to a chemical reaction of the cement and water.
    - ***Aggregates***: typically account for 60-75 percent of the total volume of a concrete mixture, described as both fine and coarse, where:
        - ***Fine aggregates***: sand and/or fine gravels with diameters less than 3/8";
        - ***Coarse aggregates*** large gravel and/or crushed stone approximately ¾" to 1" in diameter.
- **Hydration:** The chemical reaction between cement and water through which concrete hardens. Concrete does NOT harden by drying.
- **Modulus of Rupture ($f_r$)**: The stress that induces failure in concrete during flexure. Typically slightly larger than the tensile strength due to the stress distribution across the bending-member cross-section. A key property for designing concrete beams.
- **Slump:** A measure of the consistency of concrete (related to *workability*). Its units are inches. Low slump means stiff consistency, while large slump would indicate a thin consistency.
- **Tensile Strength ($f_t$)**: Less commonly used than compressive strength, as concrete is much weaker in tension (approximately 10% the compressive strength), which is why concrete structural members are typically reinforced with steel. The tensile strength is measured indirectly via a split cylinder test.
- **Water-Cement Ratio (w/c):** The most critical factor in determining a concrete mixture's strength and workability. As the w/c ratio is lowered, strength is increased, but workability is decreased.
- **Workability:** A concrete is said to be workable if it is easily transported, placed, compacted and finished without any segregation (i.e., separation of components). ASTM (American Society for

Testing and Materials) defines it as "*that property determining the effort required to manipulate a freshly mixed quantity of concrete with minimum loss of homogeneity*".

## Theory

Concrete mix design is conducted based on the required final properties of the concrete, including **compressive strength (*f'c*)**, **workability** (measured via **slump**), and applicable durability criteria (e.g., freeze-thaw, chemical exposure, sulfate resistance, permeability). The American Concrete Institute (ACI) publishes codes that establish criteria for concrete that govern mix design. Criteria may be found in both the "Standard Practice for Selecting Proportions for Normal, Heavyweight, and Mass Concrete" (ACI 211.1) and the "Building Code Requirements for Structural Concrete" (ACI 318).

The basic procedure for designing the proportions of a concrete mixture involves nine (9) steps:

**Step 1:** Select the desired value of slump for concrete workability (ACI 211.1, Section 6.3.1).

Depending on the intended application, the desired workability/slump of the concrete will be different. For example, concrete that must be pumped to upper levels of a building for placement may require greater workability than concrete that can be placed directly using the chute on the ready-mix truck. Recommended slumps for various types of construction from ACI 211.1 are shown in **Table 2.1**.

**Table 2.1: Recommended slumps for various types of construction, from ACI 211.1.**

| Types of Construction | Slump (in) | |
|---|---|---|
| | Maximum | Minimum |
| Reinforced foundation walls and footings | 3 | 1 |
| Plain footings, caissons, and substructure walls | 3 | 1 |
| Beams and reinforced walls | 4 | 1 |
| Building columns | 4 | 1 |
| Pavements and slabs | 3 | 1 |
| Mass concrete | 2 | 1 |

**Step 2:** Select the maximum aggregate size for the concrete (ACI 318-14 and ACI 211.1).

The same criteria for maximum aggregate size are presented in both ACI 211.1, section 6.3.2 and ACI 318-14 section 26.4.2.1. The nominal maximum size of aggregate used in concrete must not be greater than any of the following:

1. One-fifth (1/5) of the narrowest dimension between the sides of formwork (for example, a 5 inch thick concrete wall requires a nominal maximum aggregate size of 1 inch or less).
2. One-third (1/3) of the depth of slabs (for example, a 3 inch thick slab requires nominal maximum aggregate size of 1 inch or less).
3. Three-fourths (3/4) of the minimum clear spacing between reinforcing elements (for example, 4 inch clear spacing between reinforcing elements requires nominal maximum aggregate size of 3 inches or less).

**Step 3:** Estimate the required quantity of mix water and air content for the concrete (ACI 211.1, Section 6.3.3).

The amount of mix water and air in the concrete mix is directly linked to concrete workability (slump) and nominal maximum size of the aggregate. Higher slumps require more mix water (or other admixtures) to achieve. The addition of air entrainment to concrete also tends to increase slump; therefore air-entrained concrete mixes typically contain less water for the same slump than non-air-entrained concrete mixes. The amount of mix water is determined based on criteria in ACI 211.1 section 6.3.3. **Table 2.2** shows the relationship between slump, nominal maximum aggregate size, water, and air content for concrete from ACI 211.1.

**Table 2.2: Approximate mixing water and air content requirements for different slumps and nominal maximum aggregate sizes, adapted from ACI 211.1.**

| | Slump (in) | Water in concrete (lb/yd$^3$), for indicated nominal maximum aggregate size | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 3/8 in | 1/2 in | 3/4 in | 1 in | 1.5 in | 2 in | 3 in | 6 in |
| Non-air-entrained concrete | 1 – 2 | 350 | 335 | 315 | 300 | 275 | 260 | 220 | 190 |
| | 3 – 4 | 385 | 365 | 340 | 325 | 300 | 285 | 245 | 210 |
| | 6 – 7 | 410 | 385 | 360 | 340 | 315 | 300 | 270 | -- |
| | > 7 | -- | -- | -- | -- | -- | -- | -- | -- |
| Air-entrained concrete | 1 – 2 | 305 | 295 | 280 | 270 | 250 | 240 | 205 | 180 |
| | 3 – 4 | 340 | 325 | 305 | 295 | 275 | 265 | 225 | 200 |
| | 6 – 7 | 365 | 345 | 325 | 310 | 290 | 280 | 260 | -- |
| | > 7 | -- | -- | -- | -- | -- | -- | -- | -- |

The amount of entrained air required in the concrete mix is dictated by freezing and thawing durability criteria for concrete, as outlined in ACI 318-14, section 19.3.1. ACI 318-14 outlines four (4) classes, F0 through F3, for freeze-thaw conditions as detailed in **Table 2.3**.

**Table 2.3: Freeze-thaw class and conditions for concrete, adapted from ACI 318-14.**

| Class | Condition |
|---|---|
| F0 | Concrete not exposed to freezing-and-thawing cycles |
| F1 | Concrete exposed to freezing-and-thawing cycles with limited exposure to water |
| F2 | Concrete exposed to freezing-and-thawing cycles with frequent exposure to water |
| F3 | Concrete exposed to freezing-and-thawing cycles with frequent exposure to water and exposure to deicing chemicals |

Depending on the freeze-thaw exposure class, ACI 318-14, Section 19.3.3 dictates the target air content in the concrete based on nominal maximum aggregate size, along with maximum w/c ratio and minimum $f'_c$, as shown in **Table 2.4**.

**Table 2.4: Total air content for concrete exposed to freeze-thaw, adapted from ACI 318-14.**

| Nominal Maximum Aggregate Size (in) | Target Air Content (%) | | |
|---|---|---|---|
| | **F0** | **F1** | **F2 and F3** |
| 3/8 | NA | 6 | 7.5 |
| 1/2 | NA | 5.5 | 7 |
| 3/4 | NA | 5 | 6 |
| 1 | NA | 4.5 | 6 |
| 1.5 | NA | 4.5 | 5.5 |
| 2 | NA | 4 | 5 |
| 3 | NA | 3.5 | 4.5 |
| **Maximum w/c:** | N/A | 0.55 | F2 = 0.45<br>F3= 0.4 |
| **Minimum $f'_c$ (psi):** | 2,500 | 3,500 | F2 = 4,500<br>F3 = 5,000 |

**Step 4:** Select the water-cement (w/c) ratio for the concrete (ACI 318-14 and ACI 211.1).

The **w/c ratio** for concrete plays a significant role in the properties of concrete. For example, high w/c ratio means the mixed concrete will be more workable (higher slump), while the resulting hardened concrete will have lower **compressive strength ($f'_c$)** compared to concrete mixed with a lower w/c ratio. Section 19.2 of ACI 318-14 specifies acceptable bounds for the compressive strength of concrete, as shown in **Table 2.5**.

**Table 2.5: Limiting values for $f'_c$ based on concrete application, after ACI 318-14.**

| Application | Concrete | Minimum $f'_c$ (psi) | Maximum $f'_c$ (psi) |
|---|---|---|---|
| General | Normalweight and Lightweight | 2,500 | None |
| Special moment frames and special structural walls | Normalweight | 3,000 | None |
| | Lightweight | 3,000 | 5,000 |

To select the w/c ratio for a concrete mix, the required $f'_c$ for the hardened concrete must be established within the ranges of values listed in Table 2.5. Once a value for $f'_c$ is selected, the w/c ratio for the concrete mix can be estimated according to section 6.3.4 of ACI 211.1, which contains estimates of concrete 28-day compressive strength based on w/c ratio, as shown in **Table 2.6**.

**Table 2.6: Relationship between w/c ratio and average compressive strength of concrete, after ACI 211.1.**

| Compressive Strength at 28 Days (psi) | Water-Cement Ratio, by weight | |
|---|---|---|
| | Non-Air-Entrained Concrete | Air-Entrained Concrete |
| 6,000 | 0.41 | -- |
| 5,000 | 0.48 | 0.40 |
| 4,000 | 0.57 | 0.48 |
| 3,000 | 0.68 | 0.69 |
| 2,000 | 0.82 | 0.74 |

Section 6.3.4 of ACI 211.1 also specifies maximum permissible w/c ratio for specific types of structures, as shown in **Table 2.7**. Common values for w/c ratio range from 0.4 to 0.5.

**Table 2.7: Maximum permissible w/c ratio for concrete in severe exposure, after ACI 211.1.**

| Type of Structure | Structure wet continuously or frequently and exposed to freezing and thawing | Structure exposed to sea water or sulfates |
|---|---|---|
| This sections (railings, curbs, sills, ledges, ornamental work) and sections with less than 1 inch cover over steel | 0.45 | 0.40 |
| All other structures | 0.5 | 0.45 |

**Step 5:** Determine the amount of cement for the concrete (ACI 211.1, Section 6.3.5).

Once the amount of water and the w/c ratio for the concrete are established (Steps 3 and 4), the quantity of cement is calculated directly:

$$\frac{w}{c} = \frac{Weight\ of\ Water}{Weight\ of\ Cement} \quad \rightarrow \quad Weight\ of\ Cement = \frac{Weight\ of\ Water}{\frac{w}{c}}$$

**Step 6:** Estimate the coarse aggregate content for the concrete (ACI 211.1, Section 6.3.6).

For comparable workability of concrete, the volume of coarse aggregate in the mix is directly dependent upon the nominal maximum size for the aggregate, and the fineness modulus of the fine aggregate. ACI 211.1 provides guidance for selecting the volume fraction of coarse aggregate (i.e., the volume of coarse aggregate per unit volume of concrete), as shown in **Table 2.8**.

**Table 2.8: Volume of coarse aggregate per unit volume of concrete based on nominal maximum size and fineness modulus of the fine aggregate, after ACI 211.1.**

| Nominal Maximum Size of Aggregate (in) | Volume of oven-dry-rodded coarse aggregate per unit volume of concrete for different values of fineness modulus of the fine aggregate | | | |
|---|---|---|---|---|
| | **2.40** | **2.60** | **2.80** | **3.00** |
| 3/8 | 0.50 | 0.48 | 0.46 | 0.44 |
| 1/2 | 0.59 | 0.57 | 0.55 | 0.53 |
| 3/4 | 0.66 | 0.64 | 0.62 | 0.60 |
| 1 | 0.71 | 0.69 | 0.67 | 0.65 |
| 1.5 | 0.75 | 0.73 | 0.71 | 0.69 |
| 2 | 0.78 | 0.76 | 0.74 | 0.72 |
| 3 | 0.82 | 0.80 | 0.78 | 0.76 |
| 6 | 0.87 | 0.85 | 0.83 | 0.81 |

The volume fractions for coarse aggregate in Table 2.8 assume the aggregate is oven-dry-rodded; therefore, to determine the weight of the coarse aggregate in the concrete mix, the volume fraction may be multiplied by the oven-dry-rodded unit weight for the aggregate, as determined in the lab.

**Step 7:** Estimate the fine aggregate content for the concrete (ACI 211.1, Section 6.3.7).

After steps 1-6 have been completed, the quantity of all ingredients in the concrete is known except for the fine aggregate. The amount of fine aggregate can be estimated either by weight, or for greater accuracy, by volume.

For the method based on weight, the total weight of all other ingredients per unit volume of concrete is determined (cement + water + coarse aggregate). Then, based on the unit weight of the concrete, the amount of fine aggregate is calculated. The unit weight of concrete may be assumed based on experience with the specific materials being used. If unit weight information for the concrete is unknown, **Table 2.9** may be used for a first estimate.

**Table 2.9: First estimate of the unit weight of fresh concrete, after ACI 211.1.**

| Nominal Maximum Size of aggregate (in) | First estimate of concrete unit weight (lb/yd$^3$) | |
|---|---|---|
| | **Non-Air-Entrained Concrete** | **Air-Entrained Concrete** |
| 3/8 | 3,840 | 3,710 |
| 1/2 | 3,890 | 3,760 |
| 3/4 | 3,960 | 3,840 |
| 1 | 4,010 | 3,850 |
| 1.5 | 4,070 | 3,910 |
| 2 | 4,120 | 3,950 |
| 3 | 4,200 | 4,040 |
| 6 | 4,260 | 4,110 |

It is more accurate to determine the amount of fine aggregate to mix in concrete on a volume basis, but this requires knowledge of the volumes of water, air, cementitious materials and coarse

aggregate. For any concrete ingredient, its volume is equal to its weight divided by its (the material's) unit weight. Therefore, the unit weight of all ingredients must be accurately known to use this method.

The volume of fine aggregate is the difference between the volume of concrete and the sum of the volumes of all other ingredients.

**Step 8:** Adjust for aggregate moisture content (ACI 211.1, Section 6.3.8).

The quantities of aggregates determined in previous steps were based on oven-dry conditions; however, natural aggregates used in concrete mixes typically contain moisture – either absorbed moisture, or absorbed moisture and surface moisture. The amount of moisture in the aggregate must be accounted for when generating the concrete mix. For example, the weight of moist aggregate will be higher than for oven-dry aggregate for the same number of particles (same volume), and the weight of aggregate required per unit volume of concrete would be higher with moist aggregate than oven-dry aggregate.

In addition, moisture in the aggregate influences the amount of water to add to the concrete. Any surface moisture on the aggregate (i.e., total moisture content minus absorption) is available to react with the cement. Therefore, the amount of mix water to add should be reduced accordingly.

**Step 9:** Trial batch and test the concrete, adjust the mix design as necessary to achieve desired results (ACI 211.1, Section 6.3.9).

Once the initial concrete mix proportions have been selected, a trial batch is produced. The trial batch is then checked for unit weight, yield, air content, and workability (slump). Cylinders are also formed to confirm $f'_c$. Based on the results of the testing, the concrete mix proportions are adjusted and subsequent trial batches are produced until the desired properties are achieved.

**Other factors to consider:**

The design of a concrete mix will also be influenced by other durability considerations (other than freeze-thaw, which was considered in step 3), such as those resulting from exposure to sulfate and water, as well as for corrosion protection of reinforcement. ACI 318-14, Section 19.3.1 defines a series of exposure classes:

- S0 through S3 for sulfate exposure
- W0 through W1 for concrete in contact with water
- C0 through C2 for corrosion protection of reinforcement.

Based on the exposure, ACI 318-14, Section 19.3.2 specifies requirements for concrete such as maximum w/c ratio, minimum $f'_c$, cement type, and others. It is important that these factors be considered for appropriate concrete mix design.

**Concrete Mix Design Example:**

Given:

A reinforced concrete footing is 6 ft by 6 ft in plan view, and 2 ft thick. The minimum spacing between reinforcing bars is 1.5 inches, and the design compressive strength of the concrete is 4,000psi. The concrete is exposed to freeze-thaw cycles, but is not regularly exposed to water. For workability, the target slump for the footing concrete is 2 inches. The fine aggregate source for the concrete mix has

a fineness modulus of 2.8, an absorption capacity of 0.8% and a moisture content of 0.13%. The coarse aggregate has an absorption capacity of 0.7% and a moisture content of 0.17%. All aggregates have a dry bulk specific gravity of 2.60. The dry-rodded unit weight of the coarse aggregate is 95 lb/ft$^3$.

Design a preliminary concrete mix for this application.

**Step 1:** Desired slump = 2 inches (given).

**Step 2:** Nominal maximum aggregate size.

Criteria (1) regarding formwork and (3) regarding reinforcement spacing are applicable

1. One-fifth (1/5) of the narrowest dimension between the sides of formwork.
    - Footing is 2 ft thick, this is minimum distance between formwork.

$$\frac{1}{5}(24in) = 4.8\ inches$$

3. Three-fourths (3/4) of the minimum clear spacing between reinforcing elements.
    - Minimum reinforcement spacing = 1.5 inches

$$\frac{3}{4}(1.5\ in) = 1.125\ inches$$

Reinforcement spacing governs nominal maximum aggregate size → use 1 inch.

**Step 3:** Estimate the required quantity of mix water and air content.

Air entrained concrete is required for freeze-thaw. Since the concrete has little to no exposure to water, it is class F1. For 1 inch nominal maximum aggregate size, this requires 4.5% air content, a maximum w/c ratio of 0.55, and a minimum $f'_c$ of 3,500 psi (no issue for $f'_c$ since the design requirement is for 4,000 psi concrete).

For air-entrained concrete with 2 inch slump and 1 inch nominal maximum aggregate size, start with 270lb of mix water per cubic yard of concrete.

**Step 4:** Select the water-cement (w/c) ratio.

For air entrained concrete with $f'_c$ = 4,000psi, select an initial w/c ratio of 0.48. This is in compliance with freeze-thaw criteria of w/c < 0.55.

**Step 5:** Determine the amount of cement.

Based on the w/c ratio, the quantity of cement is computed as follows:

$$Weight\ of\ Cement = \frac{Weight\ of\ Water}{\frac{w}{c}} = \frac{270\frac{lb}{yd^3}}{0.48} = 562.5\frac{lb}{yd^3}$$

**Step 6:** Estimate the coarse aggregate content.

Fineness modulus of fine aggregate = 2.8 (given). Based on this fineness modulus and 1 inch nominal maximum aggregate size, the volume of coarse aggregate should be 0.67yd$^3$ per cubic yard of concrete mix. Weight of coarse aggregate is computed as follows:

$$Weight = Volume(Unit\ Weight)$$

$$Weight = (0.67yd^3)\left(27\frac{ft^3}{yd^3}\right)\left(95\frac{lb}{ft^3}\right) = 1{,}718.6lb\ per\ yd^3\ of\ concrete$$

**Step 7:** Estimate the fine aggregate content.

Using the volume method, the volumes of the ingredients per cubic yard of concrete are as follows:

$$Water: \quad \frac{270lb}{62.4pcf} = 4.33ft^3\left(\frac{1yd^3}{27ft^3}\right) = 0.160yd^3$$

$$Cement: \quad \frac{562.5lb}{3.15(62.4pcf)} = 2.86ft^3\left(\frac{1yd^3}{27ft^3}\right) = 0.106yd^3$$

$$Air\ (4.5\%): \quad 0.045(1yd^3) = 0.045yd^3$$

$$Coarse\ Aggregate\ (bulk/particles): \quad \frac{1{,}718.6lb}{2.60(62.4pcf)} = 10.59ft^3\left(\frac{1yd^3}{27ft^3}\right) = 0.392yd^3$$

*Note: Table 2.8 presents the oven-dry-rodded volume of coarse aggregate; we compute the bulk volume for the coarse aggregate here because the other concrete ingredients (fine aggregate, cement, water) will fill the "voids" between coarse aggregate particles which are included in the oven-dry-rodded volume.*

Total volume, less fine aggregate:

$$0.160yd^3 + 0.106yd^3 + 0.045 + 0.392yd^3 = 0.703yd^3$$

Volume of fine aggregate:

$$1yd^3 - 0.703yd^3 = 0.297yd^3\ \ fine\ aggregate\ per\ yd^3\ concrete$$

Weight of fine aggregate:

$$Weight = (0.297yd^3)\left(27\frac{ft^3}{yd^3}\right)(2.60)\left(62.4\frac{lb}{ft^3}\right) = 1{,}301lb\ per\ yd^3\ of\ concrete$$

Using the weight method, the first estimate of the unit weight of concrete with 1 inch nominal maximum aggregate size is 3,850lb/yd$^3$. The total weight of other ingredients is as follows:

$$270lb + 562.5lb + 1{,}718.6lb = 2{,}551lb$$

This implies the weight of fine aggregate in the mix is as follows:

$$3{,}850\frac{lb}{yd^3} - 2{,}551\frac{lb}{yd^3} = 1{,}299\ lb\ fine\ aggregate\ per\ yd^3\ conrete$$

Note that the weight of fine aggregate from the simplistic weight method using the "first estimate" of the unit weight of the concrete is very comparable to the weight of fine aggregate using the volume method in this case. This may not be true depending on the bulk specific

gravities of the aggregates. Assuming all of the information is known (specific gravity of materials, dry-rodded unit weight), the volume method is more accurate.

**Step 8:** Adjust for aggregate moisture content.

In this case, the aggregate moisture content is less than the absorption capacity, so water could be drawn into the aggregate pores instead of reacting with cement. However, ACI 211.1, Section 6.3.8 suggests that if absorption is less than about 1%, there shouldn't be much loss of water to absorption during initial set (i.e., mix water shouldn't need to be increased or reduced to account for absorption). If the moisture content were greater than the absorption capacity, then the quantity of mix water would need to be adjusted based on the surface moisture in the aggregate available to react with the cement.

While the mix water does not need to be adjusted in this case, since the aggregates contain moisture, the weight of aggregates for the mix must be corrected to reflect the moisture content.

Weight of coarse aggregate, dry = 1,718.6 lb

$$Water\ Weight\ (W_w) \rightarrow Moisture\ Content\ (MC) = \frac{W_w}{W_{OD}} \rightarrow W_w = MC(W_{OD})$$

$$W_w = 0.0017(1{,}718.6lb) = 2.92lb\ \ of\ water\ in\ coarse\ aggregate$$

Weight of coarse aggregate, wet:

$$1{,}718.6lb + 2.92lb = 1{,}721.5lb$$

Weight of fine aggregate, dry = 1,301 lb

$$W_w = MC(W_{OD}) = 0.0013(1{,}301lb) = 1.69lb\ of\ water\ in\ fine\ aggregate$$

Weight of fine aggregate, wet:

$$1{,}301lb + 1.69lb = 1{,}302.7lb$$

**Step 9:** Trial batch and test the concrete (not carried out).

**Summary/Final Preliminary Mix Design:**

The final weights and volumes of all ingredients to mix 1yd$^3$ of concrete for the footing are summarized in **Table 2.10.**

**Table 2.10: Final preliminary concrete mix design for the footing, including both weight and volume of ingredients per cubic yard of concrete.**

| Nominal max aggregate size = 1 inch | **Volume (yd$^3$)** | **Weight (lb) (Dry Aggregate)** | **Moist Weight (lb)** |
|---|---|---|---|
| Water | 0.160 | 270 | 270 |
| Cement | 0.106 | 562.5 | 562.5 |
| Coarse Aggregate | 0.392 | 1,718.6 | 1,721.5 |
| Fine Aggregate | 0.297 | 1,301 | 1,302.7 |
| Air | 0.045 | 0 | 0 |
| **TOTAL** | 1.000 | 3,852.1 | 3,856.7 |

## Apparatus

This lab test consists of the following materials:

1. **Concrete ingredients**: Portland cement (Type I), fine aggregate, coarse aggregate, water supply.
2. **Digital scales**: To measure the weight of each ingredient. Use larger, floor-model scales to weigh aggregates and smaller, table-top scales for water and cement.
3. **Mixing Tools**: Shovels, trowels, scoops, straightedges, buckers, mixing bowls and pans.
4. **Cylindrical Molds**: The molds must be nonabsorbent and watertight, with a smooth and/or greasy inner surface. We generally use two types:
    a. ***Small plastic molds*** which are 3” in diameter and 6” in height.
    b. ***Larger plastic molds*** which are 6” in diameter and 12” in height.
5. **Tamping Rod**: A round, straight steel rod, 5/8” diameter, approximately 24”-36” long, having one end rounded to a hemispherical tip the same diameter as the rod. This is a standard piece of concrete mixing equipment.
6. **Slump Cone**: A metal mold, which is in the form of the lateral surface of the frustum of a cone with the 8” base in diameter, the top 4” diameter, and the height 12”. The base and the top are open and parallel to each other and at right angles to the axis of the cone.
7. **Ruler**: Used to measure slump.
8. **Specimen extraction tools**: Hammer, extraction tool, etc.
9. **Curing Chamber:** An environment chamber capable of maintaining a temperature in the range of 23°C +/- 2°C with a relative humidity of at least 95%.
10. **Camera (*not provided*):** To take pictures of the process during mixing and slump test.
11. While not used in this lab, students are advised to look at both the compression machine, and the splitting tension machine to see how the concrete cylinders formed in this lab will later be tested. The TA can explain the equipment’s function.

## Procedure

You will use the following procedure to mix concrete, check the slump, and form specimens for later strength testing.

*Note: Mixing concrete can be relatively messy. Take care to maintain a neat workspace while gathering all necessary tools and ingredients.*

### *Part 1 – Gather Mix Design*

All groups will mix their concrete ingredients according to the mix design provided. The total volume of the concrete mix must be at least 0.25 $ft^3$ in order to create the specimens.

The w/c ratio will be between 0.4 and 0.65, and each lab group will be assigned a different w/c ratio within that range for their concrete mix. When the concrete is tested, strength data will be shared among the lab groups such that an evaluation of the influence of w/c ratio on concrete properties can be evaluated.

1. Verify the mix design that you have been assigned. Each group may have a different w/c ratio.

2. Select four buckets/containers (one each for cement, water, fine aggregate and coarse aggregate). Be sure to weigh them empty, such that the weight of the bucket is taken into account when measuring ingredients.
3. Measure the appropriate amounts of materials in each bucket/container. Determine the weight of each component in the concrete mix.
4. Use the w/c ratio assigned to your group and the weight of the cement in the design mix to determine the amount of water required for the mix, and fill the fourth bucket/container with the correct amount of water for the concrete mix.
5. Gather mixing tools.

### *Part 2 – Mixing Fresh Concrete*

6. Place approximately 1/3 of the fine aggregate and 1/3 of the cement into a large mixing pan and mix manually. Ensure good blending (e.g., 10 seconds of mixing is *not* sufficient).
7. Add 1/3 of the coarse aggregate and mix.
8. Add 1/3 of the water and mix thoroughly.
9. Repeat steps 6, 7, and 8 until you have everything mixed.

### *Part 3 – Slump Test (ASTM C143)*

10. Dampen the slump mold and place it on a flat, moist, nonabsorbent, rigid surface. Hold the mold firmly in place during filling by standing on the two foot pieces. Fill the mold with fresh concrete in three layers, each approximately one-third the *volume* of the mold. Note that based on the conical shape, one-third of the volume of the slump mold fills it to a depth of 2 5/8”, whereas two-thirds of the volume fills it to a depth of 6 1/8”.
11. Rod each layer with 25 strokes of the tamping rod. Uniformly distribute the strokes over the cross section of each layer. Rod the bottom layer throughout its depth. For this layer, incline the rod slightly making approximately half of the strokes near the perimeter, and then progress with vertical strokes spirally toward the center. Rod the second layer and top layers each throughout its depth, so that the strokes just penetrate to the top of the underlying layer.
12. While filling the top layer, heap the concrete above the mold before rodding is started. If the rodding induces the concrete to subside below the top edge of the mold, add additional concrete to keep an excess of concrete above the top of the mold, at all times. After the top layer has been rodded, strike off the surface of the concrete by means of a screening and rolling motion of the tamping rod.
13. Remove the mold immediately from the concrete by raising it carefully in the vertical direction. Raise the mold a distance of 12” in 5 seconds (+/- 2 sec) by a steady upward lift with no lateral or torsional motion.

    ***Note: The slump test must be completed within an elapsed time of 2-½ min from the start of the filling through removal of the mold, without interruption.***

14. Immediately measure the slump as follows:
    a. Position the slump cone on its smaller end, close to the concrete sample.
    b. Place a straight edge on the larger cone end, so as to extend over the subsided concrete.
    c. Using a ruler, determine the vertical difference between the top of the mold (i.e., the straightedge) and the displaced original center of the top surface of the specimen.

*Note:* If one side or portion of the subsided concrete shows falling away or shearing off of material, disregard the test and make a new test from another portion of the sample.

Additionally, if two consecutive tests on a sample of concrete show a falling away or shearing off of a portion of the concrete from the mass of the specimen, the concrete probably lacks necessary plasticity and cohesiveness for the slump test to be applicable.

15. Record the slump to the nearest ¼ inch.
16. Share your slump results and water content (w/c ratio) to the other groups (or provide it to the instructor to be distributed).

### *Part 4 – Preparing Sample Cylinders (ASTM C31, ASTM C192)*

You will prepare eight (8), 3-inch by 6-inch cylindrical concrete specimens, for a total volume of concrete of 0.2 $ft^3$. If you mixed at least 0.25 $ft^3$ of concrete, you should have enough to completely fill the cylinders.

17. Take a plastic cylinder mold and determine and record its weight on a digital scale.
18. Place the plastic mold on a level, rigid surface as close as possible to the location of the scale and the location where the concrete specimens will initially be cured.
19. Place the concrete in the mold in three equal layers using a scoop or a trowel. Rod each layer with the rounded end of the tamping rod using a total of 25 strokes for each layer. Rod each layer so that the tamping rod just penetrates to the top of the underlying layer. Distribute the strokes uniformly over the cross section of the mold, and for each upper layer allow the rod to penetrate about 1 in. into the lower layer. Tap the outside of the cylinder 10 to 15 times after each layer is rodded making sure to eliminate any visible voids. Make sure that the third layer contains enough concrete to fill the mold after rodding.
20. After consolidation by rodding, finish the top surface by striking it off with the tamping rod with a screening and rolling motion.
21. Gently move the filled cylinder mold to the digital scale and measure and record the weight of the filled concrete cylinder (weight of mold + concrete).
22. To prevent water evaporation from the fresh concrete, cover the specimens immediately after weighing with a non-absorptive, durable, impervious plastic sheet. Specimens may be gently moved to an initial storage place for intial curing.
23. Repeat this process to form all required specimens.
24. Make sure that all formed cylinders are labeled with a specimen number, your group number and lab section.
25. Calculate the unit weight of the wet concrete in each of your specimens. The unit weight is equal to the weight of the concrete divided by the volume of the cylinder mold, or:

$$\gamma = \frac{(Weight\ of\ Mold + Concrete) - (Weight\ of\ Mold)}{Volume\ of\ Mold}$$

### *Part 5 – Clean Up*

26. Dispose of any leftover concrete according to your instructor's directions, and thoroughly clean tools and mixing pans. **DO NOT** dispose of any concrete products down the drains.
27. Write your lab report (after concrete testing), addressing all items in the Analysis section.

### *Part 6 – Specimen Extraction & Curing*

28. After the concrete specimens have been sealed and cured in their molds for 24 hours, remove six (6) of the concrete cylinders from their respective formworks using the extraction tools. Take care not to damage any surfaces of the specimens, especially any load-bearing surfaces (i.e., the ends of the cylinders for compression testing). Leave two (2) specimens in the molds. These specimens will not be cured in the controlled curing environment.
29. As soon as each of the six specimens is removed from its mold, mark/label it and place it in the moist curing environment (curing chamber). The specimens must remain in the curing environment for the remainder of the 28 days. The curing chamber should be set to maintain a temperature of 23°C +/- 2°C with a relative humidity greater than 95%.

## Analysis

This lab will be combined with the concrete testing lab (Lab #5) into a single report. Key data to record from this lab include:

1. A table with the details of your concrete mix design.
2. A table of your w/c ratio and slump, along with the other class group results (distributed by the instructor). Plot the results (e.g., workability vs. w/c).
3. A table reporting the wet unit weight of the concrete for each specimen.

## To Include in Results/Analysis/Discussion

1. A description of the workability of your concrete mix.
2. The trend in workability versus water content.
3. A prediction of strengths.

## Reflection/Questions

1. Report your group's slump measurements and compare them with standard concrete applications (e.g., would this mix be OK for beams and columns? Foundations?) Justify.
2. Why are the molds filled incrementally, instead of all at once?
3. Type I cement was used in the mix. How many days should you wait to test the concrete? Why is this time recommended?
4. Develop a preliminary concrete mix design for a 4 inch thick basement floor slab (interior) that is not exposed to water, sulfate, or freeze-thaw. The target slump is 2 inches and the minimum design compressive strength of the concrete is 4,000psi. The aggregates have the following properties:
    - Coarse aggregate:
        - Dry-rodded unit weight = 93 pcf
        - Bulk specific gravity = 2.65
        - Absorption = 0.6%

    - Moisture Content = 1%
- Fine aggregate:
    - Fineness modulus = 2.6
    - Bulk specific gravity = 2.63
    - Absorption = 0.8%
    - Moisture content = 1.2%

Report the quantities of ingredients by weight and volume required per cubic yard of concrete.

## References

American Concrete Institute (ACI) 211.1-91 (Standard Practice for Selecting Proportions for Normal, Heavyweight, and Mass Concrete, Reapproved 2009)

American Concrete Institute (ACI) 318-14 (Building Code Requirements for Structural Concrete)

ASTM C143/C143M-15a (Standard Test Method for Slump of Hydraulic-Cement Concrete)

ASTM C31/C31M-19 (Standard Practice for Making and Curing Concrete Test Specimens in the Field)

ASTM C138/C138M-17a (Standard Test Method for Density (Unit Weight), Yield, and Air Content (Gravimetric) of Concrete)

ASTM C192/C192M-19 (Standard Practice for Making and Curing Concrete Tests Specimens in the Laboratory)

# Laboratory #3: Fracture Analysis

# Laboratory #3

Fracture Analysis

## Objectives and Overview

We have seen in Lab #1 that materials can fail in tension in either a brittle or ductile manner. However, this type of failure rarely occurs – most structural components never reach their ultimate strength in tension, even if they are designed to undertake uniaxial loads!

No material is perfect. Every material has structural flaws or defects, whether they be at the microscopic or macroscopic scale.[1] In fact, the larger a component gets, the higher the probability that there will be flaws, and that the flaws will be larger (which makes sense). This is one of the reasons we tend to use small specimens in the lab: the small size limits the chances of large flaws, and we can carefully inspect the specimens.

So how does a material fail when there is a defect? The most common mode of failure is ***fracture***.

Consider a crack in a metal specimen. If the crack was not there, stress is carried across the specimen uniformly. Unfortunately, if there is a crack (a microscopic "gap" in the system), the material cannot transfer stress across the crack. However, the stress *has* to go somewhere (based on the principles of equilibrium). As such, the stress goes *around* the crack. These stresses prefer to take the shortest path, so generally there is a lot of stress building up at the edge or tip of the crack. This causes a huge stress concentration at the crack tip. If the local stress is high enough, the material will fail, and the crack gets a little bit longer. As a result, the stress path is extended, and the stress concentration forms again. This results in repeating the process, and the crack continues to propagate until there is no more material to break – *i.e.*, fracture.

It may or may not be obvious that the larger the crack length, the more stress must be redirected around the crack tip. As a result, the stress that initiates fracture is dependent on (a) the length of the crack, (b) the resistance to fracture of the material (so-called fracture toughness), and (c) the system geometry.

Since failure in fracture is so sudden and catastrophic, fracture testing is one of the most critical tests for a material. However, because it is geometry dependent, it is also difficult to replicate the stress and crack conditions. Here, we must simplify the situation, mimicking uniaxial tension tests, but introducing defects/cracks. Moreover, we are looking at linear elastic fracture conditions (so-called linear fracture mechanics, or LFM).

During this lab, you will perform tension tests on both metallic and plastic samples. Testing will be conducted on specimens of varying crack lengths to illustrate the relationship to fracture. The instrument used for the test records force-displacement (F-Δ) data that will allow us to plot stress-strain (σ-ε) curves for each specimen, giving us insight into the mechanical behavior of the material. Like the uniaxial tension test, a series of specimens will be loaded in tension up to fracture, and thus the test is known as a destructive test (you cannot use the specimens more than once!). Here, analysis of the material behavior will be used to determine mode I fracture toughness ($K_{IC}$).

[1] This is universally true due to the laws of thermodynamics. No material can be perfect. Ever.

## Definitions

- **Crack Behavior**: Fracture toughness is a quantitative way of expressing a material's resistance to brittle fracture when a crack is present. If a material has high fracture toughness it will probably undergo ductile fracture. Brittle fracture is very characteristic of materials with low fracture toughness.
- **Griffith Criterion**: Fracture mechanics was developed during World War I by English aeronautical engineer, A. A. Griffith in order to explain the failure of brittle materials. Griffith introduced an artificial flaw in his experimental glass specimens. The artificial flaw was in the form of a surface crack that was much larger than other flaws in a specimen. The experiments showed that the product of the square root of the flaw length ($a$) and the stress at fracture ($\sigma_f$) was nearly constant, which is expressed by the equation:

  $$\sigma_f \sqrt{a} \approx \text{constant}$$

  Griffith developed a thermodynamic approach to explain the relation that he observed. The growth of a crack requires the creation of two new surfaces and hence an increase in the surface energy. If one was to compute the change in the free energy (surface energy − elastic energy) as a function of the crack length, failure occurs when the free energy attains a peak value at a critical crack length, beyond which the free energy decreases by increasing the crack length (i.e., by causing fracture).
- **Fracture Mode:** There are three ways of applying a force to enable a crack to propagate
  - **Mode I**: a normal tensile stress perpendicular to the crack (e.g., crack opening)
  - **Mode II**: a shear stress along the crack length (e.g., crack sliding)
  - **Mode III**: a shear stress normal to the crack length (e.g., crack tearing).
- **Fracture Strength ($\sigma_f$):** The stress on a material at which fracture occurs.
- **Fracture Toughness ($K_{IC}$):** A property that describes the ability of a material containing a crack to resist fracture. The linear-elastic fracture toughness of a material is determined from the stress intensity factor (K) at which a thin crack in the material begins to grow. It is denoted $K_{IC}$ and has the units of $Pa * \sqrt{m}$ or $psi * \sqrt{in}$. The subscript I denotes mode I crack opening.

  $K_{IC}$ is the critical K at the critical stress at the critical distance. That is, $K_{IC}$ is the stress intensity factor when failure occurs ($\sigma_f$) as $r \to 0$. Analytically, this is a difficult parameter to determine.
- **Stress Intensity:** Another a method of calculating the amount of energy available for fracture is in terms of the asymptotic stress and displacement fields around a crack front in a linear elastic solid. This expression for the stress field around a crack tip is:

  $$\sigma = \left(\frac{K}{\sqrt{2\pi r}}\right) f(\theta)$$

  where $\sigma$ is the stress, $r$ is the distance from the crack tip, θ is the angle with respect to the plane of the crack, and $f$ is a function that depends on the crack geometry and loading conditions. The quantity $K$ is called the stress intensity factor, which is material dependent.

## Theory

**The required theory/data analysis for this lab includes:**

**(1) Griffith Criteria:**

Griffith found an expression for the constant that he observed in terms of the surface energy of the crack by solving the elasticity problem of a finite crack in an elastic plate. He determine the following equation for the fracture stress $\sigma_f$:

$$\sigma_f = \sqrt{\frac{2E\gamma}{\pi a}}$$

where *E* is the Young's modulus of the material and *γ* is the surface energy density.

**(2) Stress Intensity Factor & Fracture Toughness:**

From **stress intensity**:

$$\sigma = \left(\frac{K}{\sqrt{2\pi r}}\right) f(\theta)$$

Solving for the critical K:

$$K_{IC} \approx \lim_{r \to 0} \frac{\sigma}{f(\theta)} \sqrt{2\pi r}$$

For the current specimen geometry,

$$K_{IC} \approx 1.64 \sigma_f \sqrt{\pi a}$$

Thus, if the failure stress and crack length are known, the fracture toughness can be approximated. To improve the estimation, it is better to fit the toughness to a linear regression rather than rely on a single calculation. Rearranging the above equation yields:

$$\sigma_f = \frac{K_{IC}}{1.64\sqrt{\pi a}}$$

Thus, the slope of a stress versus $a^{-1/2}$ plot can be used to determine fracture toughness.

## Apparatus

This lab test consists of the following materials:

1. **Tension machine (ADMET UTS):** Your lab instructor will explain how the compression/tension machine works. In general, a specimen is stretched at a prescribed displacement rate (e.g., displacement-controlled) and the load is recorded until the specimen fails (how is "failure" defined in this case?).
2. **Digital calipers:** For measuring the length and diameter of the specimens.
3. **Digital scales**: For measuring the weight of the specimens.
4. **Two (2) material specimens:** (1) steel and (1) PVA. The specimens are rectangular. They will have a crack on one side.
5. **Camera (*not provided*):** To take pictures of the specimens during testing and post-failure.

## Procedure

You will use the following procedure to determine the fracture toughness of the materials.

1. Put the specimen into the grips on the ADMET machine. The grips should be flat to accommodate the rectangular cross-section.
2. Measure the dimensions of the specimen, including the crack length, $a$. Record your data.
3. Verify the test settings, including displacement rate and failure conditions. Record these conditions. The displacement rate should be set to 1 inch/minute for the testing.
4. Start the test.
5. Readings of applied force and deformation (elongation) are shown on the computer monitor, as well as the force-deformation curve.
6. At least one group member is responsible for taking pictures prior to the test, during the test, and after failure. These pictures should be included in the lab report as support for your analysis.
7. After failure, remove the specimen from the grips and inspect it.
8. The data acquisition system attached to the testing machine writes records of applied force and deformation at regular intervals to a file. Students will be given that file of force-deformation records (the format is delimited text, which can be opened in Excel or Matlab).
9. Repeat steps 1-8 for each material specimen.
10. Carry out the analysis described below.
11. Write your lab report, addressing all items in the Analysis section.

## Analysis

Your lab reports must contain the following:

1. Convert the computer readings (force-deformation) of the metallic and plastic specimens to stress-strain, and develop stress-strain plots for the two specimens (total of 2 plots).
2. On your plots, identify all of the important values.
3. Supply the fracture strength and crack length to the TA. The TA will distribute all the values to the class.
4. Plot failure strength versus crack length for the two materials (two plots). Fit with a linear regression and determine fracture toughness.
5. Using Griffith's criterion, approximate the surface energy for (a) your test data and (b) the classes' test data. Comment on any differences.
6. Determine the specific fracture toughness (i.e., normalize by density).

## To Include in Results/Analysis/Discussion

1. Discuss and compare the failure of the tensile specimens. Which material has a higher fracture toughness?
2. Which material has the ability to absorb the most energy *prior to fracture?*

## Reflection/Questions

1. According to the Griffith criteria, what is the strength of a perfect material (i.e., when $a \to 0$)? Does this make sense?
2. According to the stress intensity factor equation, what is the stress at the crack tip (i.e., when $r \to 0$)? Does this make any sense?
3. Considering questions (1) and (2), what is the strength of a material with an extremely small defect (i.e., both $a \to 0$ and $r \to 0$)?

# Laboratory #4: Flaw Tolerance

# Laboratory #4

Flaw Tolerance

## Objectives and Overview

We have previously seen that defect or crack size is detrimental to the strength of a material (**Lab #3**). Moreover, we have also mentioned that no material is perfect; defects exist at some scale. However, some materials can withstand many more defects than others, and still achieve near ideal strength behavior. Such materials are known as flaw tolerant or damage tolerant. **Flaw Tolerance** (or damage tolerance) is a property of a material (or a structure) relating to its ability to sustain defects and still withstand some threshold of load. For example, how many "pieces" can be broken and still achieve 80% ultimate strength. Clearly, a material that can sustain ten distributed cracks and still carry load is more flaw tolerant than a material that fails with one small chip.

The approach to materials and engineering design to account for flaw tolerance is based on the assumption that flaws can and do exist in any structure and such flaws propagate/increase with usage; for example, crack extension due to fatigue (multiple load cycles), accidental damage, exposure, and corrosion. Flaw tolerance can be defined as a state of material in which pre-existing cracks (from various sources) do not propagate even as the material is loaded to failure near its limiting strength. Like fracture, the scale of the damage comes into play; every material reaches an ultimate point of failure if the defect is large enough.

As it turns out, many biological materials such as bone and nacre are flaw/damage tolerant. This is a good thing, as you do not want your bones breaking in two at the slightest crack (it takes quite a load to break a bone!). Materials scientists have discovered a few tricks that nature plays to achieve flaw tolerant behavior:

1. ***Composites***: Materials such as bone and nacre consist of mineral platelets and protein-based materials in a "brick and mortar" like structure. The mineral phase is what gives the material its strength. However, the mineral phase is typically very brittle, while the protein phase is typically very ductile. Cracks like to propagate through brittle materials, but when they hit the protein, they slow down, change direction, and dissipate a lot of energy. If the material is arranged correctly, the cracks can only extend as far as the mineral "bricks."
2. ***Amorphous Materials***: Protein-based materials are extremely similar to polymer-based materials (e.g., plastics, rubbers). They do not have a well-defined "crystal" structure like metals and ceramics. As such, it is more difficult for a crack to extend, as there is no easy "plane" to make fracture surfaces. However, such amorphous materials are typically very soft and weak under pure tension.
3. ***Boundaries***: In pure crystalline materials, cracks can extend extremely quickly if there is nothing to stop them. However, many metals have so-called grain boundaries, where the crystallographic orientation of the metal changes direction. This crystal "mismatch" creates a boundary of high energy due to the mismatched atoms, which is more difficult for the crack to pass through. As a general rule, the more grain boundaries (and thus smaller grains), the higher the flaw tolerance as well as ultimate strength.

For all of the reasons described above, the molecular structure or geometry is what effectively creates flaw tolerance. Many efforts are currently being undertaken to find the best geometries for composite

materials, engineered grain boundaries, and control of polymer phases to increase the flaw tolerance and strength of high performance materials.

Here, we are going to explore a type of common household material – tape – in order to investigate the concept of flaw tolerance and loss of strength. Through simple tension tests and the introduction of random "flaws", we can easily deduce the best applications for particular tape materials.

## Definitions

- **Flaw Measure:** Some metric for determining severity of flaws. For example: number of flaws, total combined crack length, volume of corroded voids, rust area, etc.
- **Flaw Tolerance:** The ability of a material to sustain damage with nominal loss of material strength. In terms of fracture mechanics, the ability of a material to halt crack propagation and eliminate stress concentrations.
- **Reduced Strength Factor:** The ratio of material strength with flaws to ideal material strength. The reduced strength factor is a function of the magnitude of flaws, where:

$$\beta(n) = \frac{\sigma(n)}{\sigma_0}$$

  where "*n*" is the flaw measure, $\sigma(n)$ the strength with flaws, and $\sigma_0$ the ideal strength (no defects). As $\beta \rightarrow 1$, the material is flaw tolerant.

## Theory

**No new theory is required for this exercise beyond the principals of stress and strain (Lab #1) and fracture (Lab #3).**

## Apparatus

This lab test consists of the following materials:

1. **Tension machine (ADMET UTS):** Your lab instructor will explain how the compression/tension machine works. In general, a specimen is stretched at a prescribed displacement rate (e.g., displacement-controlled) and the load is recorded until the specimen fails (how is "failure" defined in this case?).
2. **Digital calipers:** For measuring the length and diameter of the specimens
3. **Official "flaw applicator" (a nail).**
4. **Three (3) material types:** (1) duct tape, (2) electrical tape, (3) packing tape. Rolls of tape will be provided. They will be generic tapes purchased from a local hardware store.
5. **Camera (*not provided*):** To take pictures of the specimens during testing and post-failure.

## Procedure

You will use the following procedure to evaluate the flaw tolerance of different tapes:

1. Make three (3) samples of each type of tape (9 samples in total) by:
    a. Measuring approximately 300 mm of tape and cutting the length.
    b. Folding the tape over on itself (carefully, such that there are no wrinkles and the tape is aligned as accurately as possible by hand), such that you have an approximately 150 mm specimen that is tape on both surfaces (i.e., no sticky side exposed).
    c. Measure the width and thickness of the tape with the calipers.
    d. For 25 mm on both ends, apply some additional tape for the machine to grip (the TA will demonstrate). Therefore, the "free length" of tape will be approximately 100 mm. Measure with the calipers and confirm.
2. Verify the test settings, including displacement rate and failure conditions. Record these conditions. The displacement rate should be 1 inch/minute for the test.
3. At least one group member is responsible for taking pictures prior to the test, during the test, and after failure. These pictures should be included in the lab report as support for your analysis.
4. Test one of each type of tape without flaws by:
    a. Putting the specimen into the grips. The grips should be flat to accommodate the thin rectangular cross-section.
    b. Start the test.
    c. Readings of applied force and deformation (elongation) are shown on the computer monitor, as well as the force-deformation curve.
    d. After failure, remove the specimen from the grips and inspect.
    e. Save the file of force-deformation records (the format is delimited text which can be opened in Excel or Matlab).
5. Apply one defect to one specimen of each type of tape. Repeat the test procedure (step 4).
6. Apply five to ten defects (*n*) on one specimen of each type of tape (your TA will direct you how many defects to use; for a particular type of tape, each lab group should use a different number of defects). Repeat the test procedure (step 4).
7. Carry out the analysis described below.
8. Write your lab report, addressing all items in the Analysis section.

## Analysis

Your lab reports must contain the following:

1. Convert the computer readings (force-deformation) of the specimens to stress-strain, and develop stress-strain plots for the three materials. Plot the defect results on the same axis for each material (total of 3 plots with 9 data sets).

2. On your plots, identify all of the important values.
3. Determine the reduced strength factor for 1 defect and *n* defects.
4. Attain the reduced strength factors from the other groups (TA will provide).
5. Plot reduced strength factors for each material as a function of *n* on the same axis.
6. Determine the toughness of each material type (recall, toughness = area under the stress-strain curve) and number of flaws. Plot the toughness versus *n*. Only use your group's data for this (you will have three points to plot – no flaws, 1 flaw and *n* flaws for a particular type of tape).

## To Include in Results/Analysis/Discussion

1. Discuss and compare the failure of the tapes. Comment on:
    a. Which tape has the highest strength without defects? With defects?
    b. Which tape has the highest toughness without defects? With defects?
    c. Which tape is the most flaw tolerant?
2. Do the mechanical properties of each tape justify their uses/intended applications? Discuss.

## Reflection/Questions

1. Do the molecular and/or microstructures of concrete and steel have features that could increase flaw tolerance? Describe.
2. What practical application would knowing the critical flaw size or concentration enable?
3. Is flaw tolerance always a good thing? Describe how a material with ideal flaw tolerance would fail. Describe a case where flaw tolerance may be disadvantageous.

# Laboratory #5: Concrete Testing

# Laboratory #5

Concrete Testing

## Objectives and Overview

During this lab, concrete specimens made during Laboratory #2 will be tested for (1) compressive strength using the compression machine, and (2) tensile strength using the compression machine.

When testing hardened concrete, testing may be performed using destructive or nondestructive methods to determine the specimen's compressive strength ($f_c'$). Commonly, this is the ***only*** property that is directly measured. Standard equations have been developed to approximate other mechanical properties (such as Young's modulus or tensile strength) based on compressive strength. We will perform tests to enable some comparisons to these standard equations.

Structural engineers typically design concrete mixes to be used in reinforced concrete buildings based on a specified compressive strength; however, once the concrete mix is hardened, the resulting compressive strength must be checked against the design strength in order to ensure that the strength requirements have been met. For example, concrete may be tested for its compressive strength by physically crushing a specimen. Unfortunately, this option may not always be feasible since we are often interested in the actual strength of a hardened concrete structure (such as a dam or a foundation wall) that cannot be damaged or destroyed. It is difficult to guarantee that test specimens are subject to the same conditions as large structures.

Here, the instrument used for the test records force-displacement (F-δ) data that will allow us to plot stress-strain (σ-ε) curves for each specimen, giving us insight into the mechanical behavior of the concrete.

## Definitions

- **Compressive strength ($f_c$):** The stress on a concrete specimen under the maximum compressive force/load.
- **28-Day Compressive Strength ($f_c'$)**: The stress on a concrete specimen under the maximum compressive force/load after 28-days of curing. This is the industry standard for concrete strength (and the "prime" designation).

  ***Note: It should be noted that compressive strength is not an intrinsic property of concrete and values are dependent on many factors, such as specimen size, mixing procedure, curing procedure, and age. Therefore, results must be interpreted carefully.***

- **Tensile Strength ($f_t$)**: The maximum attainable stress on a concrete specimen subjected to direct tensile forces/loads. Determined indirectly via a split tension test due to intrinsic difficulties in testing concrete in uniaxial tension (due to premature crushing at grips).

## Theory

**The required theory/data analysis for this lab includes:**

**(1) Force-displacement to stress-strain, and $f_c'$**

Stress is calculated by dividing the tensile force applied to the specimen by the original area of the specimen, or:

$$\sigma = \frac{F}{A_0}$$

where $\sigma$ is the stress (psi or MPa), $F$ is the load carried by the specimen during tension test (lb or N), and $A_0$ is the initial or original cross-sectional area of the specimen (in$^2$ or m$^2$).

Strain is calculated by dividing the change specimen length ($\Delta L = L - L_0$) by the original specimen length ($L_0$). However, the test equipment records the displacement (δ). The new specimen length is thus:

$$L = L_0 + \delta$$

Such that the strain is:

$$\varepsilon = \frac{\Delta L}{L_0} = \frac{L - L_0}{L_0} = \frac{(L_0 + \delta) - L_0}{L_0} = \frac{\delta}{L_0}$$

where $\varepsilon$ is the strain (in/in or m/m; e.g., unitless), $\delta$ the recorded machine displacement (in or m), and $L_0$ the initial or original gage length (in or m)

Thus, the output force-displacement data can be easily converted to stress-strain data by recording the initial dimensions of the specimen.

For $f_c'$, stress is calculated by dividing the maximum compressive force applied to the specimen by the original area of the specimen, which has a standard diameter, $D_0$, such that:

$$f_c' = \frac{F_{max}}{A_0} = \frac{F_{max}}{\frac{1}{4}\pi D_0^2} = \frac{4F_{max}}{\pi D_0^2}$$

**(2) The determination of Young's modulus *via* linear regression**

For linear elastic materials, stress is related to strain via Hooke's law, where:

$$\sigma = E\varepsilon$$

Note that this is just a linear equation, $y = Mx + b$, where y is stress, x is strain, the intercept is zero and the slope is Youngs' modulus, $E$. Note that this is equivalent to defining the modulus as:

$$E = \frac{d\sigma}{d\varepsilon}$$

Thus, to determine the modulus, one must simply fit a linear trendline to the stress-strain data.

Choose a section of the data that looks linear (typically for strains less than 2% or 3%, but is dependent on the material; use your own judgement!), and fit a trendline to it using Excel or Matlab.

Moreover, it is good practice to calculate the $R^2$ value, which will give an indication of the goodness of fit.

**(3) Determining tensile strength from split tension test**

Due to difficulty in applying uniaxial tension to a concrete specimen, the tensile strength of concrete is commonly determined via an indirect test method known as the split cylinder or split tension test.

A compressive load is applied pointwise to a cylinder laid on its axis. This results in a tensile splitting stress induced down the face of the specimen. The tensile strength is then calculated as:

$$f_t = \frac{2F_{max}}{\pi D_0 L_0}$$

where $F_{max}$ is the maximum recorded compressive load, $D_0$ the diameter of the cylinder, and $L_0$ the length of the cylinder.

Note while force-displacement data will be recorded during the split tension tests, only $F_{max}$ is necessary, as the displacement ***does not*** relate to strain.

**(4) Abrams' Law**

Abrams' law states that in concrete materials, for a mixture of workable consistency, the strength of concrete is determined by the ratio of water to cement. As the water content increases, the strength decreases. Many studies have shown that the relationship between water content (specifically water/cement ratio) and strength is an exponential function, where:

$$f_c' = \frac{A}{B^{w/c}}$$

Where w/c is the water/cement ratio, and "A" and "B" are fitted empirical constants. A given batch plant, with consistent aggregates, cement supply, water supply, and mixing conditions (such as humidity, temperature) will yield unique "A" and "B" values. Once these constants are known, the strength of the concrete mix is predictable, and the w/c can be adjusted to yield desired values without a complete mix redesign.

**(5) Effect of Age on Compressive Strength**

If the concrete is tested earlier than 28-days, $f_c'$ can be approximated by using the approximate fraction of 28 day compressive strength values shown in **Table 5.1**.

**Table 5.1: Approximate fraction of 28 day concrete compressive strength for concrete that is less than 28 days old.**

| Age (*days*) | % of 28-day strength attained |
|---|---|
| 7 | 70% |
| 14 | 80% |
| 21 | 90% |

**(6) Empirical Relationships for $f_c'$**

A long history of concrete testing and experimental correlation has led to a set of standard equations to approximate concrete properties if $f_c'$ is known. Assuming normal strength, normal density concrete, the ACI code prescribes (in units of psi):

$$E = 57{,}000\sqrt{f_c'}$$

$$f_t = 6.7\sqrt{f_c'}$$

$$f_r = 7.5\sqrt{f_c'}$$

Note that, even though each of the above equations takes the square root of $f_c'$, the resulting units is still considered *psi*. These are empirical equations!

**(7) ASTM C39 Types of Failure**

Standard descriptors are used to describe concrete cylinder failure when subject to compression (**Figure 5.1**). Note that most failure modes are at an incline, indicating shear-dominated failure.

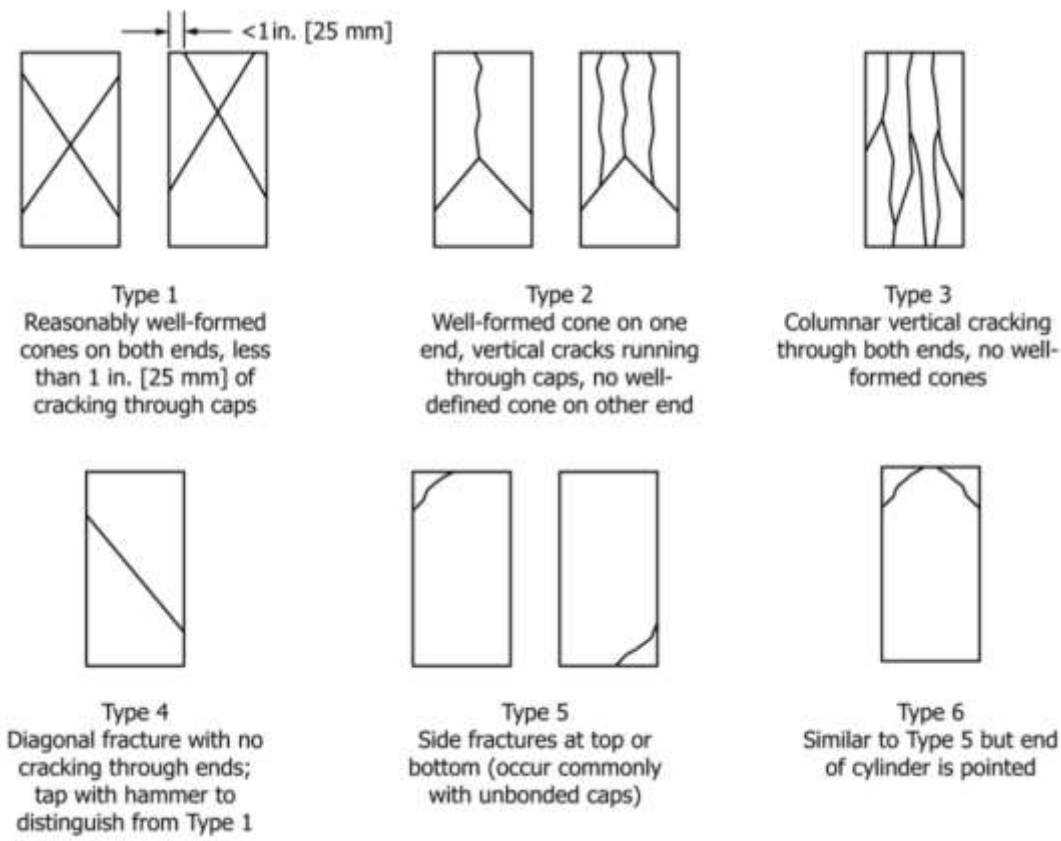



**Figure 5.1: Typical configurations and descriptions of concrete cylinder failure when tested in compression from ASTM C39.**

## Apparatus

This lab test consists of the following materials:

1. **Concrete specimens**: Eight (8) cylinder specimens prepared in Laboratory #2, five (5) for compressive stress testing and three (3) for split tension testing.

2. **Digital scale:** To measure the weight of each concrete cylinder.
3. **Calipers and measuring tools:** To measure the dimensions of the concrete cylinder after extraction from the mold and curing.
4. **Compression machine:** Your lab instructor will explain how the compression machine works. Records force-displacement data.
5. **Camera** (*not provided*): To take pictures of the specimens before, during, and after testing.

## Procedure

You will use the following procedure to test concrete. Note that the tests need not be performed in a specific order. Work amongst the groups to schedule efficient testing.

1. Make sure the compression machine is set up and properly connected to the computer and LabView software for the collection of force and displacement data. The TA and/or lab technician will assist with this step.
2. Remove the concrete cylinders from the curing environment. Do this as close as possible to the time when the specimens will be tested – if there will be a wait, it is best not to remove all of your cylinders at once.
3. Inspect the specimens for any apparent flaws (e.g., cracks, honeycombing). Make notes.
4. Using the calipers and measuring tools, measure and record the diameter and height of each specimen. Take 3 measurements/readings of each to determine average diameter and height. Compute the volume of the specimen based on its dimensions.
5. Measure and record the weight of the cured concrete specimen on a digital scale. Determine the unit weight of the concrete specimen after curing. The unit weight is equal to the weight of the concrete divided by the volume of the concrete, or:

$$\gamma = \frac{Weight\ of\ Concrete}{Volume\ of\ Concrete}$$

6. Load specimens as soon as possible. Cylinders must be moist when tested. Surface drying of the moist-cured specimens results in changes in the expected compressive strength. Note the conditions.
7. Before placing the concrete cylinder in the testing machine, clean the faces of the cylinder and bearing blocks. Carefully align the axis of the cylinder with the center of thrust of the spherically seated (top) block.

   Note:

   The testing machine shall have sufficient loading capacity to fail the concrete cylinders, and shall provide the proper loading rate. The testing machine shall be equipped with two steel bearing blocks with hardened faces: one of which is a spherically seated block (top block) that will bear on the upper surface of the specimen, and a second solid block (bottom block) on which the specimen shall rest. Minimum dimensions of bearing faces of the blocks shall be at least 3% greater than the diameter of the specimen tested. Faces of the two steel blocks shall be smooth, except for concentric circles, which shall be inscribed to facilitate proper centering of concrete cylinders.

8. Zero the load on the compression machine.
9. Start the testing machine. The top steel block will move downwards towards the spherically seated block.
10. As soon as the top surface of the specimen initiates contact with the spherically seated block, make sure that uniform seating is obtained. As a result of seating, a small load may be observed on the compression machine display.

    Note:

    For specimens that do not have a level surface, use neoprene caps (rubber disks) between the surface of the block and concrete cylinder – both top and bottom.

11. Zero the load on the display on top of the compression machine, and zero the force and displacement readings in the LabView data collection software.
12. Load the specimen. The rate of loading shall be within the range of 35 (+/- 7) psi per second. The rate of loading corresponds to the rate of movement of the top block with respect to the bottom block. The load should be applied continuously and without shock, until the concrete cylinder fails. A rate of loading exceeding the prescribed rate is permitted during the application of the first half of the loading phase. Use the prescribed rate of loading during the latter half of the loading phase. Do not adjust the loading rate until the specimen begins yielding quickly, just prior to failure.
13. Readings of applied force and deformation (elongation) can be seen on the computer monitor, as well as the force-deformation curve. The data acquisition system attached to the testing machine also writes records of applied force and deformation at regular intervals to a file.
14. Record the maximum load carried by the specimen during the test. Also note the w/c ratio associated with each specimen. Save the data from the test in LabView using a descriptive name (lab group number, w/c ratio, lab date) The data from all groups at all different w/c ratios will be shared with the whole class. The force-deformation data files are in delimited text format, which can be opened in Excel or Matlab.
15. At least one group member is responsible for taking pictures prior to the test, during the test, and after failure. These pictures should be included in the lab report as support for your analysis.
16. Note the prevalence of cracking on the specimen as the load is increased up to failure, as well as the type of failure that occurs (see **Figure 5.1**).
17. Repeat steps 1-14 for each compression specimen (5 total; 3 from the moist curing environment, 2 that were not cured in a controlled curing environment).
18. Repeat the loading process (steps 1-14) for the split tension test, with the cylinder laid on its side instead of loaded axially. A total of 3 moist-cured cylinders should be tested in split tension.
19. Carry out the analysis described below.
20. Write your lab report, addressing all items in the Analysis section.

## Analysis

Your lab reports must contain the following:

1. A summary of Laboratory #2, including the mix design used, the workability analysis, and the concrete

unit weight.

2. A comparison of the unit weight of the wet concrete to the unit weight of the cured concrete.
3. A plot of the stress-strain data for compressive tests on moist-cured specimens on a common axis, indicating $f_c'$.
4. Adjust for 28-day strength (if necessary).
5. For the plots in item #3, fit trendlines for Young's modulus. Use the standard formula to compare the experimental value with the prediction.
6. A plot of the stress-strain data for compressive tests on specimens cured in an uncontrolled environment, on a common axis, indicating $f_c'$.
7. Description of failure modes.
8. Calculation of the tensile strength from split tension test (plot not required). Use the standard formula to compare the experimental value with the prediction.
9. Collect other groups' strength values for moist cured specimens and plot $f_c'$ vs. w/c.
10. Use Abrams' Law and determine the constants "A" and "B". Plot Abrams' Law on the same axis as the experimental data. Comment on the fit.

## To Include in Results/Analysis/Discussion

1. Describe the trend in workability versus water content.
2. Comment on the accuracy of the standard equations using $f_c'$ to predict other values.
3. Compare the unit weight of the wet concrete to the unit weight of the cured concrete. Comment on any differences.
4. Compare the compressive strength of the concrete based on the moist-cured specimens (cured in the curing chamber) vs. the two specimens cured in an uncontrolled environment.
5. Is Abrams' Law reasonable for Snell Engineering's Materials Lab? Why or why not?
6. List some sources of error and variability in the lab, considering both the mixing and testing aspects (e.g., how do the unit weights of your test specimens compare?). Do you think testing concrete is as accurate as other materials' tests? Why or why not?

## Reflection/Questions

1. Why are the concrete molds filled incrementally, instead of all at once?
2. What differences do you observe in mechanical properties between the cylinders that were moist-cured and those that were not? Why are the properties different?

## References

ASTM C39/C39M-18 (Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens)

# Appendix A: Lab Report Requirements

## General:

Lab reports are a type of technical writing, in which clarity, conciseness, and logic are primary goals. Leave flowery writing, bias, and opinions for poetry and short stories.

Lab reports are intended to convey three critical pieces of information to the reader:

(1) what material property is desired to be tested/characterized/defined;

(2) what laboratory test or experiment was performed to achieve this, and;

(3) the results of that laboratory test or experiment.

Note that when listing what material property is to be tested, you must assume that the reader has *zero* knowledge of the material or relevant theory. Thus all pertinent details and key equations must be given. This does not mean you have to "dumb down" the material and explain the fundamentals from basic principles, but merely that all *necessary* details and formulations must be given and/or addressed.

Moreover, as most labs involve the gathering of numerical data and/or test results, care should be taken in the presentation of such data in the form of plots, graphs, diagrams, and tables. Such items should be able to be understood *independently* from the written text of the report.

Someone who wishes to use your lab report to source material data should be able to understand each result quickly.

***Remember, at the end, a lab report should be able to be used as a guide to replicate the lab exactly - and achieve near-equivalent results – by someone who has zero knowledge of the topic.***

## Submission:

All lab reports are submitted with a coversheet the details the lab day and time, lab group, lab group members, and their individual contributions to the overall lab report. The coversheet is included in Appendix B along with a lab report grading rubric.

## Requirements:

Key components of the lab report include:

1. A cover page, including abstract
2. Introduction
3. Theory/Procedure/Methods
4. Results, Analysis & Discussion
5. Reflection/Questions*
6. Conclusion
7. Appendices

*In a strict technical lab report, the section entitled "Reflection/Questions" may be omitted. Reflection here involves extrapolating the observed results to engineering applications, comparison to lecture material, etc. Questions will guide your thought process. The required content of section 5 will be given for each lab.

***The following pages are structured to demonstrate the lab report format/contents.***

CIVE 2261 – Materials Laboratory

SEMESTER – YEAR

Professor

Lab/Experiment #

**A Descriptive Lab Title**

*Report Date*

***Abstract:***

The abstract is a concise summary of your report. It should be about eight to ten lines long and single-spaced, or approximately 150-300 words. Write for the intended audience (think of your high school science teacher), with the goal of helping the reader decide whether the full report is worth reading. The abstract should explain the purpose and scope of the report and summarize the key findings. When summarizing findings, don't write, "results are presented," leaving your reader wondering what the results were. Instead, describe the results, or state what was interesting about the results. Abstracts are standalone documents, and *never* refer to figures or acronyms that are found in the report. Most writers find it best to write the abstract after completing the report.

Submitted By: Name, name, name

Experiment Completed: Date

### *Introduction*

This section has two parts: a brief introduction and objectives (each one warrants its own paragraph).

First, in a short paragraph, introduce the lab. Answer the simple questions: What is the material? What is the test? What fundamental phenomena is being observed? What desired property are you attempting to measure?

Second, explain the key objectives or hypothesis of the lab. You can discuss the theory that the laboratory test/experiment is based on and the principles that were used in the test/experiment, but leave details (e.g., equations or formulas) for the next section.

### *Theory/Procedure/Methods*

This section should provide all the "technical details" for the lab experiment, as well as the analysis (e.g., if you use a certain math equation in the results to achieve a plot, it should be described here).

First, expand on any theory that will be used. List necessary equations. Motivate any unknown variables that will be tested/measured.

Second, inform the reader as to the procedure you followed *step-by-step*. Enumerate the steps if there are many. Include key apparatus used in the experiment. You are not expected to describe the apparatus or procedures in great detail. Mention any specific, standardized tests that were done (e.g., ASTM C143 for concrete slump).

Third, describe the results/data/output that you expected the laboratory test/experiment to produce, prior to testing/experimentation, relating it to the theory behind the experiment.

### *Results, Analysis & Discussion*

In this section you will present your results and explain their meaning. In the first paragraph inform the reader of main components in this section and what they should expect to read in the coming paragraphs.

Present and analyze your results using tables, graphs and figures as necessary. **This section *must* include:**

(1) *at least* one **table**; and/or

(2) *at least* one **graph/plot** illustrating your data or results derived from the data;

(3) *at least* one **figure** illustrating the test specimen or laboratory test/experiment.

Think carefully about alternative ways to best illustrate the information so that your reader may clearly understand what you are trying to present, and so that the underlying meaning of the data is apparent. If you have data from other groups, it should appear in this section. Determine the best way of comparing data and identify any variation or consistencies. Ask yourself pertinent questions - for example, when there are multiple tests/experiments, is it meaningful to show a total? Or perhaps it is more meaningful to calculate an average, a difference, or a standard deviation? Carefully think about how to best present your results, asking yourself what is important to the reader, what kind of graphic is most effective (or dramatic) in demonstrating your point, and what information belongs in prose (text) versus a table or figure. Be sure to incorporate your tables and graphs in your discussion – number and label them, and refer to them as necessary.

The analysis should be consistent with the purpose of the laboratory test/experiment (as described in the first section of the report). It should compare the results with those expected (also described in the first section of the report).

Each lab assignment may include some analysis items or issues to address. These should be addressed in this section where you – the writer – feel most appropriate. Each issue should have its own paragraph. If an analysis issue merits multiple paragraphs, please provide a sub-heading for that issue.

This whole section should be written as a cohesive discussion – not a piecewise listing of results/data.

### *Reflection/Questions*

Reflection should be a few sentences to a few short paragraphs in length. Such a section would not normally be part of a technical report, however, we include it because it is a valuable part of the learning process.

Critically evaluate this lab and your results. Point out and discuss any possible sources of error. Offer recommendations for improving the test/experiment or lab organization and any thoughts about what you learned, or how the lab relates to the materials class. You may also include any personal conclusions or thoughts in this section. Do not speak in generalities in this section – be specific to your group.

Some lab assignments will include application questions that will be answered here. When answering the questions, rewrite the question first, and then provide the answer. All questions must be numbered and solutions must be typed.

### *Conclusion*

Your conclusion is a brief summary, highlighting key findings and any other significant aspects of the lab. New material should never be presented in a Conclusion section. If you find that, in order to support your conclusion, you are introducing new material, it is an indication that you need to revise the Results and Analysis section by including the new material there.

### *Appendices*

Appendices, if needed, should be included at the end of the report. Include any information that was collected and used in discussion and calculations, but was not relevant enough to be shown in the results and analysis section. Appendices must be referred to in the body of the report.

Example appendices are:

a) ***Data sheets***: include if raw data is not tabulated in body of the report. If you have many data sheets containing similar information, a one-page sample sheet will suffice.

b) ***Sample or extended calculations***: A sample calculation should be shown for calculations that are not obvious. For example, if you used a spreadsheet to compute values in a table in the results and analysis section, you should show how those values were generated in this section of the report. That way, if there is a number in the table that does not make sense, the reader can see how it was calculated and make comments on it. It is not necessary to provide sample calculations for common mathematical procedures such as summations or averages.

c) ***Copies of material*** from references such as procedure descriptions or standards (not usually needed).

Designate the appendices (A, B, C, etc.) and precede each with a title page.

***Final things to remember when you are writing up your lab reports:***

- Use your own words when explaining the theory, procedure, etc. Do not plagiarize (including the laboratory manual!).
- Use references ***only if necessary***. The lab report should be standalone.
- Use a consistent tense (i.e., past or present).
- Prepare neat tables and graphs. Give tables and graphs both a number and a meaningful name, e.g., "*Table 1: Displacement and Load Readings*."
- Indicate the units in tables, plots, and calculations. In a plot, always label the axes. Significant digits must be kept consistent in tables. Decimal places should line up in each column.
- Tables must be "called out" and explained in the text before you present them.
- Think about how precise your measurements are: often two decimal places are appropriate (i.e., 28.50 lb. vs. 28.4980167282 lb.).
- Construct and format your tables so that the logic of the calculations is obvious to the reader. It should be unnecessary to explicitly explain most of the calculations. If there are complex calculations that are not obvious, explain and provide sample calculations in an appendix.
- Write in a style that fits your intended audience: a high school science teacher who is not an expert in materials testing procedures, but is familiar with standard engineering and scientific terms.
- Type entire report (including formulas and calculations – nothing should be handwritten).
- Use Times New Roman (or similar), size 12.
- Single-space within paragraphs, and double-space between paragraphs/sections.
- Number pages (bottom right).

# Appendix B: Lab Report Coversheet and Grading Rubric

# CIVE 2261: Lab Report Cover Sheet

Lab Day & Time: _____

Lab Group No.: _____

## ***Group Member Names & Contribution:***

| Member | Name | Contribution/Section(s) |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |

## ***Instructions:***

1. Fill out the lab day and time and lab group number at the top of the submission cover sheet.
2. Enter all group members' names in the contribution table.
3. Enter each group member's contribution in the contribution table (e.g., Results and Discussion, Abstract and Introduction).
4. Attach this cover sheet to your lab report for submission.

## ***Grading:***

| Component of Report | | Points | Score |
|---|---|---|---|
| **Abstract** | | 10 | |
| **Introduction & Theory/Procedure/Methods** | | 15 | |
| **Results** | Reasonable; analysis & discussion | 20 | |
| | Graphics and sample calculations | 20 | |
| **Reflection/Questions** | | 20 | |
| **Conclusion** | | 10 | |
| **Appendices** | | 5 | |
| | | **100** | |

# CIVE 2261: Lab Report Grading Rubric

Following is the grade table appearing on the lab report cover sheet:

## ***Grading:***

| Component of Report | | Points | Score |
|---|---|---|---|
| **Abstract** | | 10 | |
| **Introduction & Theory/Procedure/Methods** | | 15 | |
| **Results** | Reasonable; analysis & discussion | 20 | |
| | Graphics and sample calculations | 20 | |
| **Reflection/Questions** | | 20 | |
| **Conclusion** | | 10 | |
| **Appendices** | | 5 | |
| | | **100** | |

## ***Grading Rubric:***

Following is a detailed rubric indicating the breakdown of points for each component of the lab report. All lab reports must be typed/computer generated. Handwritten work will not be accepted. Documents should be well formatted. Format items to consider include fonts (heading vs. body text), spacing (line, paragraph), text justification, and page numbering.

**Abstract (10 pts total)**

The abstract is a short summary of the report, consisting of a paragraph of about 8 to 10 lines of text.

The abstract should include the following:

- *The importance of the laboratory test/experiment.* (2 pts)
- *The purpose of the report.* (4 pts)
    - A declarative statement which summarizes the specific topic and goals of the report. Should be specific and precise, not general, broad or obscure. Should be concise (one or two sentences), clear and goal oriented (stated in terms of desired outcomes).
- *The scope of the report.* (2 pts)

    - A description of what you are going to describe/address in the report, and a brief summary of key components of the report. If there is a major issue that your report will not be addressing, that should be stated here with a reason why.
- *Key findings from the laboratory test.* (2 pts)

**Introduction & Theory/Procedure/Methods (15 pts total)**

The introduction is the opening section of the body of the lab report. The introduction should include the following:

- *Objectives of the lab.* (4 pts)
    - If applicable, briefly explain the theory/principles used in the lab test here.
- *Description of test procedures.* (8 pts total)
    - Mention/reference the relevant test standard (e.g., ASTM C143 for concrete slump) (2 pts)
    - Summarize the procedure you followed in brief, using your own words. Inform the reader about the key apparatus used (you may include a picture here to help explain!) (6 pts)
- *Discussion of expected results of the lab test.* (3 pts)

**Results/Analysis/Discussion (40 pts total)**

The primary section of the lab report, where the results are communicating effectively using words, tables, figures, images, etc. The results/analysis/discussion should include the following:

- *The lab test results, analysis and discussion.* (20 pts total)
    - Results make sense and are reasonable. (5 pts)
    - Analysis and discussion addresses the meaning of the results, comparing actual results with expected results. Refer to the course textbook, lecture notes, and/or lab manual as appropriate. (15 pts)
- *Graphics and sample calculations to present the results effectively.* (20 pts total)
    - Tables and figures have a number and a meaningful name (table captions are placed above tables, figure captions are placed below figures).
    - All equations are numbered.
    - All plots have axis labels.
    - Tables and figures are appropriately sized and legible.
    - Fonts are consistent between writing/text, tables and figures.
    - Sample calculations shown for each equation/formula used.

**Reflection/Questions (20 pts total)**

In these sections of the lab report, you critically evaluate the lab and your results, and respond to questions posed in the lab manual. The reflection/questions should include the following:

- *Reflection.* (5 pts)

    - Critically evaluate your results, pointing out and discussing possible sources of error. Describe your thoughts about what you learned or how the lab relates to the materials class. You may also include personal conclusion or thoughts here, but be specific to your group, do not speak in generalities. You may also offer recommendations for improving the lab test, or lab organization here.
- *Questions.* (15 pts)
    - Rewrite the questions posed in the lab manual, and then respond to them.
    - Answer the questions completely and clearly. Use text, equations, tables or figures as appropriate, following the same criteria as described for the Results/Analysis/Discussion.

**Conclusion (10 pts total)**

The conclusion is the closing section of the body of the lab report, and should summarize the report and the key findings.

- *Brief summary.* (5 pts)
- *Highlight key findings.* (5 pts)

**Appendices (5 pts)**